Program overview

19-May-2025 12:06

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bachelor Electrical Engineering**

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
B-EE 2024	**Bachelor Electrical Engineering 2024**						
B-EE jaar 1 2024	**Bachelor Electrical Engineering Year 1 2024**						
EE1C1	Linear Circuits A	5					
EE1C2	Linear Cicuits B	5					
EE1D1	Digital Systems A	5					
EE1D2	Digital Systems B	5					
EE1E1	Electrical Energy Fundamentals	5					
EE1G1	Introduction to Electrical Engineering	5					
EE1L1	Integrated Project 1	5					
EE1L2	Integrated Project 2	5					
EE1M1	Calculus	5					
EE1M2	Calculus and Linear Algebra	5					
EE1M3	Linear Algebra and Differential Equations	5					
EE1P1	Electricity and Magnetism	5					
B-EE jaar 2 2024	**Bachelor Electrical Engineering Year 2 2024**						
EE2C1	Transistor Circuits	5					
EE2C2	Mixed-Signal Circuits and Systems	5					
EE2G1	Electrical Engineering for the Next Generation	10					
EE2L1	Integrated Project 3	5					
EE2M1	Probability and statistics	5					
EE2P1	Electromagnetics	5					
EE2P2	Semiconductor Physics and Devices	5					
EE2S1	Signals and Systems	5					
EE2S2	Systems and Control	5					
EE2T1	Telecommunication and Sensing	5					
Electives							
EEX01	Introduction to Machine Learning	5					
EEX02	Communication Networks and Algorithms	5					
EEX03	Microwave Engineering	5					
EEX04	Technologies for Energy Transition	5					
EEX05	Chip Design	5					
B-EE jaar 3 2024	**Bachelor Electrical Engineering Year 3 2024**						
EE3C11	Electronics	5					
EE3D11	Computer Architecture and Organisation	5					
EE3L11	Bachelor graduation project Electrical Engineering	15					
EE3P11	Electromagnetics	5					

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
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B-EE 2024	
Director of Studies	Dr.ir. R.C. Hendriks
Program Coordinator	C. Thepass
Contact for students	Let op! De antwoorden van veel van je vragen staan al op het web (https://www.tudelft.nl/studenten/faculteiten/ewi-studentenportal/)! Ben je eerstejaars en heb je het antwoord op jouw vraag niet op de website kunnen vinden? Neem dan als eerste contact op met je student-mentor of je docent-mentor/tutor. Studenten kunnen ook altijd terecht bij de commissaris onderwijs van EE (onderwijs-etv@tudelft.nl). Die weet zelf op veel vragen antwoord, maar kan ook doorverwijzen of bij andere mensen van de opleiding dingen navragen. Gedurende je hele opleiding worden er regelmatig opleidingsbijeenkomsten gehouden: In het eerste jaar zijn er kwartaalbijeenkomsten, in de hogere jaren semesterbijeenkomsten. De bijeenkomsten staan in het rooster. Voor vragen over hulp en begeleiding bij persoonlijke problemen of studieachterstand kunnen alle studenten contact opnemen met studieadviseurs. Check ook hier eerst de website: https://www.tudelft.nl/studenten/ewi-studentenportal/organisatie/studieadviseurs Voor antwoord op vragen over specifieke vakken kijk je eerst op de Brightspace-site van dat vak en als je het antwoord daar niet kunt vinden neem je contact op met de verantwoordelijk docent van dat vak. Wil je meedoen aan het verbeteren van de opleiding? Geef je dan op voor de student feedback panels van jouw jaar. Neem ook hiervoor contact op met je commissaris onderwijs van de ETV.
Term of validity	Let op! De informatie die je hieronder vindt is informatief van aard en niet bindend! Bindende informatie vind je in de reglementen (https://www.tudelft.nl/studenten/faculteiten/ewi-studentenportal/). Houd Brightspace in de gaten voor wijzigingen en kom naar de opleidingsbijeenkomsten voor de meest recente informatie.
Program Goals	De afgestudeerde van de opleiding Electrical Engineering: - Bezit een brede en grondige kennis van en heeft vaardigheden in de fundamentele ingenieurswetenschappen, die de basis van de elektrotechniek vormen. Hieronder worden verstaan: wiskunde, natuurkunde, informatica en systemen en modellen. De afgestudeerde kan deze kennis actief toepassen op elektrotechnische systemen en beheerst de kennis op een zodanig niveau dat toegang verkregen kan worden tot internationaal geaccrediteerde masteropleidingen Electrical Engineering. - Heeft basistechnisch-wetenschappelijke kennis van en heeft vaardigheden in de belangrijkste elektrotechnische disciplines, te weten netwerktheorie, telecommunicatietechniek en systemen, signaalbewerking, elektronische schakelingen, elektrische energietechniek, halfgeleidercomponenten en computertechniek. De afgestudeerde kan deze kennis actief toepassen voor het analyseren en ontwerpen van elektrotechnische systemen en het lezen en begrijpen van de wetenschappelijke literatuur in bovengenoemde vakgebieden. - Bezit basiskennis van en heeft vaardigheden in methodes en gereedschappen voor het modelleren, simuleren, ontwerpen en uitvoeren van experimenten en onderzoek van/aan elektrotechnische systemen. De afgestudeerde kan deze kennis actief toepassen voor het analyseren en synthetiseren van elektrotechnische systemen (op een hoog abstractieniveau). - Kan een bijdrage leveren aan het oplossen van technologische problemen door een systematische wetenschappelijke aanpak. Dit betreft de analyse, het definiëren van innovatieve oplossingen, het onderkennen van de haalbaarheid, het onderkennen en verwerven van ontbrekende kennis, evenals het onderkennen van de betrekkelijkheid en beperkingen van deze kennis en van de uitwerking van de oplossing. - Kan zowel individueel als in (multidisciplinaire en multinationale) teams werken en waar nodig initiatief nemen. - Kan effectief communiceren (waaronder presenteren en rapporteren) over zijn werk, t.a.v. informatie, problemen, ideeën en oplossingen aan zowel de professionele collegae als aan een niet-specialistisch publiek. - Is in staat om relevante informatie te verzamelen en te interpreteren en kan de technologische, bedrijfskundige, maatschappelijke en ethische gevolgen van zijn werk evalueren en de verantwoordelijkheid nemen met betrekking tot duurzaamheid, economie en sociaal welzijn. De afgestudeerde draagt bij aan de wetenschappelijke praktijk (onderzoeksysteem, relatie met opdrachtgevers, publicatiesysteem, belang van integriteit, etc.). - Toont initiatief en kan kritisch reflecteren (met ondersteuning) op het eigen denken, beslissen, en handelen en dit daarmee bijsturen. De afgestudeerde kan de eigen competenties op peil houden en uitbreiden door permanente zelfstudie, met een hoge mate van zelfstandigheid. Zie het onderwijs- en examenreglement voor de volledige tekst van de eindtermen van de opleiding.
Program Structure 1	De opleiding EE duurt drie jaar. Het eerste jaar bestaat elk uit 12 studieonderdelen van elk 5 EC: 10 vakken en twee projecten. In het tweede jaar volg je in het derde kwartaal twee vakken van 5 EC en kies je een keuzevak van 5 EC. In het vierde kwartaal volg je een 10 EC keuzeproject en een 5 EC-vak. De rest van het tweede jaar is gelijk aan het eerste jaar. Het derde jaar wijkt behoorlijk af. De eerste helft ervan volg je een minor (zie onder). Daarna zijn er nog 2 vakken van 5 EC en een keuzevak van 5 EC. Je laatste 15 EC besteed je dat jaar aan je bachelorafstudeerproject. Elk jaar is opgedeeld in 4 kwartalen van 10 weken en tijdens elk kwartaal heb je drie vakken die parallel lopen. Gedurende de 10 weken wordt er van je verwacht dat je elke week ongeveer 40 uur aan je studie besteedt. Drie vakken per kwartaal betekent ongeveer 13 uur per week aan elk van die vakken besteden. Niet al die uren zijn ingepland, je moet veel zelfstandig werken (zelfstudie).
Program Structure 2	EE-studenten en hun docenten maken deel uit van de academische gemeenschap van de TU Delft. Elke gemeenschap kent gedragsregelen en de TU Delft heeft deze vastgelegd in een ethische code (Code of Ethics). De onderstaande regels komen uit deze code: Docenten en studenten: - hebben respect voor anderen; - treden betrokken, transparant en integer op; - dragen bij aan een inspirerende werk- en studieomgeving; - vertrouwen elkaar en vermijden belangenverstrengeling; - respecteren elkaars eigendommen en middelen en die van de universiteit. Docenten: - gedragen zich rechtvaardig en respectvol ten opzichte van elkaar en van studenten; - mikken op hoge kwaliteit en streven altijd naar verbetering; Studenten: - stellen zich rechtvaardig en respectvol op ten opzichte van elkaar en van universiteitsmedewerkers; - halen het beste uit zichzelf door actief deel te nemen aan hun opleiding en extra curriculaire activiteiten; - stimuleren elkaar en hun docenten door analytische vragen te stellen en goed beargumenteerde discussies te voeren.
Exam requirements	In de exameneisen staat wanneer je in aanmerking komt voor een EE-diploma. Je vindt ze in de Regels en Richtlijnen van de Examencommissie (http://studenten.tudelft.nl/ewi/reglementen/) . Samengevat staat er dat je een EE-diploma krijgt als je alle vakken van de opleiding, inclusief de minor, hebt gehaald met als resultaat een V (voldoende) of een cijfer van 6 of hoger.

With Honours Regulation	Studenten die binnen de voor de opleiding geldende tijd (drie jaar) de BSc opleiding EE afmaken, gemiddeld voor alle vakken minimaal een 8 hebben en ook minimaal een 8 hebben voor het bachelorproject krijgen hun diploma met de aantekening met lof (cum laude). De exacte eisen kun je vinden in de Regels en Richtlijnen van de Examencommissie.
Minor Semester	In de eerste helft van je derde jaar volg je een minor. De minor is er om je de mogelijkheid te geven je kennis te verbreden of een buitenlandervaring op te doen. Meer informatie vind je bij het derde jaar.
Gives access to	De bachelor EE geeft o.a. toegang tot de EWI Masters Electrical Engineering, Sustainable Energy Technology en Computer and Embedded Systems Engineering aan de TU Delft. Ook heb je toegang tot verschillende masters aan de Universiteit Twente en Technische Universiteit Eindhoven. Met een schakelprogramma kun je toegelaten worden tot een groot aantal andere Masterprogrammas van de TU Delft en andere universiteiten. Zie http://doorstroommatrix.nl/ voor een uitgebreid overzicht.
Expected prior Knowledge	De toelatingseisen van de opleiding EE kun je vinden in hoofdstuk 2 van het Studentenstatuut. Ze bevatten in ieder geval wiskunde en natuurkunde op VWO-B niveau.
Honours track	Voor goed presterende en ambitieuze studenten is er vanaf jaar 2, een honoursprogramma waarbij zij al betrokken kunnen worden bij onderzoek. Zie https://www.tudelft.nl/studenten/ewi-studentenportal/onderwijs/honours-programme.

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B-EE jaar 1 2024	
Director of Studies	Dr.ir. R.C. Hendriks
Director of Studies	Dr.ir. N.P. van der Meijs
Program Coordinator	C. Thepass
Program Coordinator	B. Abdikivanani
Prerequisites	Let op! De Ingangseisen zijn in de OER vastgelegd. Mocht er verschil zijn tussen wat er hier staat en wat in de OER staat, dan is datgene wat in de OER staat geldig. In het eerste jaar gelden voor de volgende twee studieonderdelen ingangseisen: Je mag pas aan het onderdeel (project) EE1L1 Integrated project 1 meedoen, als je de course lab van het vak EE1C1 Linear Circuits A hebt behaald. Je mag pas aan het onderdeel (project) EE1L2 Integrated project 2 meedoen, als je het vak EE1L1 Integrated Project 1 en de course labs van de vakken EE1D1 Digital Systems A en EE1D2 Digital Systems B hebt behaald. Mentoraat Het BSc EE Y1 mentoraatprogramma loopt het hele jaar door. Registratie van dit programma is opgenomen als een pass/fail binnen de vakken EE1G1, EE1L1 en EE1L2. Rekenmachine Alleen een eenvoudige rekenmachine zonder mogelijkheden voor opslag van tekst of formules is toegestaan. Geschikte types zijn (varianten van) de Texas Instruments TI-30 en de Casio FX-82.
Introduction 1	Inschrijven Voor de (deel-)tentamens van álle vakken moet je je uiterlijk tot 14 kalenderdagen voor de tentamendag via OSIRIS inschrijven. Wil je bij nader inzien toch niet meedoen aan een studieonderdeel of tentamen waar je je voor hebt ingeschreven? Schrijf je dan ook weer uit! Zie voor uitleg https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens
Introduction 2	Zoals bij alle TU Delft bacheloropleidingen, geldt bij EE de norm van 45 EC als bindend studieadvies (BSA). Kijk voor meer informatie op http://bsa.tudelft.nl.

EE1C1	Linear Circuits A	5
Responsible Instructor	Dr.ing. I.E. Lager	
Responsible Instructor	Dr. F. Fioranelli	
Contact Hours / Week x/x/x/x	10/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	N.A.	
Expected prior knowledge	N.A.	
Course Contents	This course, in its entirety, concerns the SYSTEM interpretation and modelling of relatively complex, linear circuits. At the core of its first part (EE1C1) is the calculation of voltages and currents in circuits consisting of current and voltage sources, resistances, inductances and capacitances. The basic components and the different analysis methods for direct current (DC) are firstly introduced. The operation of capacitances and inductances will be then explained. Combining circuit analysis methods with the behaviour of inductances and capacitances will form the basis for studying first- and second-order circuits. The course will continue in its second part (EE1C2) when the acquired knowledge and skills will be applied to examining circuits operating under steady-state alternating current (AC) conditions and more elaborate transient conditions. The EE1C1 part also contains a CourseLab during which the concepts introduced and tested during the lectures and the seminars will receive a physical demonstration. During these CourseLabs, you will develop skills in assembling electronic circuits and using standard laboratory (testing) equipment. The lab sessions will be prepared during some sessions in which practical linear circuits problem will be discussed and their impact on the ensuing lab classes will be stressed.	
Study Goals	After following this course you should be able to: Circuit theory competences / skills: Express and apply the basic concepts of electrical circuits; Estimate, examine and evaluate the basic circuital quantities (current, voltage, charge, energy, power); Identify, interpret and examine the basic circuit elements: independent sources, resistances, inductances, capacitances, op-amps (strictly as linear circuit elements) and dependent sources; account for them via their specific operational/mathematical models; Express Ohm's law and Kirchhoff's theorems, and apply them for calculating currents and voltages in circuits; Identify series and parallel circuits, and apply the needed relations for calculating currents and voltages in circuits; Express and interpret the nodal and mesh methods, and apply them to circuit analysis; Express and interpret special instruments for analysing circuits: source transformation, superposition, Thévenin/Norton equivalents, and apply them to circuit analysis; Identify and interpret first-order electric circuits, and apply the suitable formalism for evaluating their transient behaviour; Identify and interpret second-order electric circuits, and apply the suitable formalism for evaluating their transient behaviour. CourseLab / skills: Interpret electronic schematics; Solder electronic components; Use basic laboratory (testing) equipment: sources, function generators, multi-meters, and oscilloscopes.	
Education Method	16 hrs lectures 32 hrs seminars 16 hrs CourseLabs 6 self-assessed graded homework assignments (SGHs) 2 partial exams	
Literature and Study Materials	Charles K. Alexander, Matthew N. O. Sadiku, Fundamentals of Electric Circuits, 7th edition, McGraw-Hill. Handouts (copies of the lecture slides) are made available after the lectures the lectures are made available. Supplementary reading offered via Brightspace.	
Assessment	This course has: 6 self-assessed graded homework assignments (SGHs), handled via a TU Delft platform the SGHs yield a sliding-weight bonus for the exam (details will be communicated at the start of the course) CourseLab preparatory + lab sessions; a successful completion of the CourseLab is needed to be passed with a sufficient for validating the exam grade 2 examinations, (weeks 1.5 and 1.10) A resit. The final grade of the course consists of the following components: Written Partial Exam 1 Written Partial Exam 2 Final grade = 0.5 * Written Partial Exam 1 + 0.5 * Written Partial Exam 2 Final grade calculation: Final grade (T) = 0.5 * Written Partial Exam 1 + 0.5 * Written Partial Exam 2 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Exam Hours	Partial exams: 2 hrs Resit: 3 hrs	
Permitted Materials during Tests	During the exam, students can use a simple non-programmable and non-graphical calculator (like the TI-30 or Casio FX-82). Students are allowed to have a handwritten A4 with their own summary / notes during the exam. The exam is NOT an open book exam. During the exam, students are allowed to ask questions to clarify the wording.	
Co-Instructor	Dr.ir. N.P. van der Meijs	
Co-Instructor	Dr. M. Ghaffarian Niasar	
Co-Instructor	K.M. Dowling	
Co-Instructor	Dr. H.A.A. Bastawrous	

EE1C2	Linear Cicuits B	5
Responsible Instructor	Dr.ing. I.E. Lager	
Responsible Instructor	Dr. F. Fioranelli	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	N.A.	
Expected prior knowledge	N.A.	
Course Contents	This course, in its entirety, concerns the SYSTEM interpretation and modelling of relatively complex, linear circuits. Upon building on the basis laid in the preceding part (EE1C1), the second part of the course (EE1C2) starts by introducing the methods for examining circuits under steady-state, alternating current (AC) operation. These techniques will then be used for: (i) performing entry-level frequency analysis; (ii) modelling (and designing) simple filters; (iii) evaluating the power transfer in AC circuits. The operation of magnetically coupled circuits will be touched upon. The course will subsequently progress towards inspecting more elaborate circuit transients via the Laplace transform. It will be concluded by briefly introducing basic two-port concepts.	
Study Goals	After following this course you should be able to: Estimate and examine circuits fed by steady-state, AC, sinusoidal signals and evaluate voltages and currents by using phasors (and phasor diagrams); Identify, classify and interpret frequency responses, transfer functions, poles, zeros; understand Bode diagrams and employ them to circuit analysis; Express and interpret the resonance phenomenon and resonant circuits; Identify, classify and interpret filter networks (low-pass, high-pass, band-pass and band-stop filters); examine and explain passive and active filters and apply them to practical applications; Express, interpret and calculate instantaneous power, average power, maximum average power, and infer the effective value concept; Estimate and examine magnetically coupled circuits, the coupling factor, and the ideal transformer; apply these concepts in the analysis of electrical circuits; Have an entry-level understanding of the direct and inverse Laplace transform; apply a table look-up + use of basic theorems strategy to calculating direct/inverse Laplace transforms; Identify and interpret initial conditions, and examine more elaborate transients by means of Laplace transform; Express and interpret the black-box concept and apply it to two-ports and their modelling; evaluate two-ports Z- and Y-parameters; account for series and parallel concatenation of two-ports.	
Education Method	16 hrs lectures 32 hrs seminars 6 self-assessed graded homework assignments (SGHs) 2 partial exams	
Literature and Study Materials	Charles K. Alexander, Matthew N. O. Sadiku, Fundamentals of Electric Circuits, 7th edition, McGraw-Hill. Handouts (copies of the lecture slides) are made available after the lectures the lectures are made available. Supplementary reading offered via Brightspace.	
Assessment	This course has: 6 self-assessed graded homework assignments (SGHs), handled via a TU Delft platform 2 examinations, (weeks 2.5 and 2.10) A resit. The result of the SGHs is expressed as B, and ranges from 0 to 10. R is calculated as the average of the results of all but one of the SGHs, in which any non-made SGH is graded with a 0; either the least-scoring SGH (when all SGHs were done) or a non-made one are omitted from the average. The results of both partial exams are averaged, this gives the exam grade E. The final course grade C is calculated as follows: C = E + B * (10-E) / 100. To pass the course, a final grade C of at least 5.76 points in scale 0-10 is needed. The final grade will be rounded to half points. The SGHs result is valid for the partial exams and the resit, but only in the year the assignments have been made. The resit includes the contents of both partial exams. The result of the partial exams expires for the resit and is not transferable to the next academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: OER Art 29, sub 4).	
Exam Hours	Partial exams: 2 hrs Resit: 3 hrs	
Permitted Materials during Tests	During the exam, students can use a simple non-programmable and non-graphical calculator (like the TI-30 or Casio FX-82). Students are allowed to have a handwritten A4 with their own summary / notes during the exam. The exam is NOT an open book exam. During the exam, students are allowed to ask questions to clarify the wording.	
Co-Instructor	Dr.ir. N.P. van der Meijs	
Co-Instructor	Dr. M. Ghaffarian Niasar	
Co-Instructor	K.M. Dowling	
Co-Instructor	Dr. H.A.A. Bastawrous	

EE1D1	Digital Systems A	5
Responsible Instructor	Dr.ir. A.J. van Genderen	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	none	
Course Contents	In this course, you will learn as student the basics of digital systems. You will learn to design, simulate, build and emulate custom and standard digital circuits. To be able to accomplish the above, you will get familiar with several hardware related topics including, among other things, the overall design of computer systems, data representation in computers, boolean algebra, logic minimization, sequential circuits, Finite State Machines, and standard modules such as multiplexers, registers, etc.. In addition, you will also get an introduction to the hardware description language SystemVerilog and use it during practical lab sessions. Topics are: - Introduction to digital design: representation, numbers - Boolean algebra - Logic minimization - Sequential circuits and FSMs - Standard modules (multiplexers, decoders, ROM, registers, counters, etc.) - Digital implementation technology (CMOS gates) - SystemVerilog: structural, behavioral, testbenches, simulation	
Study Goals	You should be able to: 1- explain the basic idea of how digital systems work and how they are built. 2- describe digital circuits by means of Boolean algebra and simplify these circuits (logic minimization). 3- design and analyze custom and standard combinatorial and sequential circuits. 4- implement Finite-State Machines. 5- describe and analyze digital circuits in CMOS technology 6- create, simulate and emulate structural and simple behavioral SystemVerilog descriptions.	
Education Method	lectures, instructions, labs	
Computer Use	QuestaSim, Quartus, espresso	
Literature and Study Materials	- Digital Design and Computer Architecture, RISC-V Edition, July 12, 2021, by Sarah Harris, David Harris (Paperback ISBN: 9780128200643, eBook ISBN: 9780128200650) - Lecture slides - Course lab manual	
Assessment	2 written partial exams, 1 re-exam, course lab assignments. In addition, there will be a test in week 4 and week 8. The result of the tests is expressed as R, ranging 0 to 10. The R figure is the average of the results of all the tests, in which a non-made test results into a 0. The results of the two partial exams are averaged. This gives the exam grade T. The final grade of C of the course is then calculated as follows: C = T + R x (10-T) / 100. This calculation is done with one decimal accuracy, and the final grade will be rounded up to half. The result of the on-line tests, R, is valid for the partial exams and the resit, but only in the year the on-line tests have been made. The final mark is valid only when all course lab assignments have been successfully completed. The resit includes the contents of both partial exams. The result of the partial exams expires for the resit and is not transferable to the next academic year. In case of insufficient results for the course labs, a repair option may exist in accordance with Article 2 Exameneisen, Clause 4, of the Uitvoeringsregeling 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	All study materials (book, slides, manuals) may be used during the exam except old exams.	
Co-Instructor	Dr.ir. M. Taouil	
Co-Instructor	H.N. Abunahla	

EE1D2	Digital Systems B	5
Responsible Instructor	Dr.ir. A.J. van Genderen	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Expected prior knowledge	EE1D1 Digital Systems A	
Course Contents	The course covers several topics related to digital circuits and systems, including combinatorial and sequential modules, the structure of a processor, and serial communication protocols. The course also covers how to design digital applications in hardware or software, as well as the hardware-software co-design space. The course concludes with a discussion of how to optimize behavioral specifications to logic circuits at the gate level using software synthesis tools. By the end of the course, students should be able to design digital circuits and applications, and optimize their designs for performance and flexibility depending on application requirements. Topics are: - Computer Arithmetic - Sequential circuit timing: tsu, thold, minimum clockperiod, clockskew, - SystemVerilog synthesis - Working principles of FPGA (LUT, FF, BRAM,...) - Serial communication protocols: I2C, SPI, UART,... - RTL systems - Basics of RISC-V instruction set, addressing modes, addressing formats, assembly - language - Simple RISC-V architecture - Basics of the C programming language	
Study Goals	You should be able to: 1- design and analyze arithmetic circuits 2- describe and use serial communication protocols such as I2C, SPI and UART. 3- describe the structure of a processor and how to program a processor in assembly language. 4- explore the hardware-software co-design space available for digital applications. 4a- design a simple application entirely in hardware (for example, if high performance is paramount). 4b- design the same application (partly) in the form of software that is running on existing hardware such as a standard microprocessor (for example, if flexibility is paramount). 5- optimize (synthesize) a behavioral specification in System Verilog to a logic circuit at the gate level using software synthesis tools.	
Education Method	lectures, labs	
Computer Use	QuestaSim, Quartus	
Literature and Study Materials	- Lecture slides - Lab manual - Digital Design and Computer Architecture, RISC-V Edition, July 12, 2021, by Sarah Harris, David Harris (Paperback ISBN: 9780128200643, eBook ISBN: 9780128200650)	
Assessment	Written exam, re-exam, course lab assignments. In addition, there will be a test in week 4 and week 8. The result of the tests is expressed as R, ranging 0 to 10. The R figure is the average of the results of all the tests, in which a non-made test results into a 0. If the (re-)exam grade is T, then the final grade of C of the course is calculated as follows: C = T + R x (10-T) / 100. This calculation is done with one decimal accuracy, and the final grade will be rounded up to half. The result of the on-line tests, R, is valid for the exam and the resit, but only in the year the on-line tests have been made. The final mark is valid only when all course lab assignments have been successfully completed. In case of insufficient results for the course labs, a repair option may exist in accordance with Article 2 Exameneisen, Clause 4, of the Uitvoeringsregeling 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr.ir. M. Taouil	
Co-Instructor	Dr.ing. M. Shahraki Moghaddam	

EE1E1	Electrical Energy Fundamentals	5
Responsible Instructor	Dr. M. Cvetkovic	
Responsible Instructor	Dr. J. Dong	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Strongly advised: EE1P1 Electricity and magnetism	
Course Contents	Starting with the energy conversion principles, this course will introduce the electrical engineering students to the critical energy conversion components and fundamentals of power systems. After the course, the students will be able to analyse and model energy components and power systems. This is essential knowledge for future engineers in the energy transition.	
Study Goals	By the end of this course, you should be able to: Electrical energy conversion components part: - Explain the power system based on AC and DC connections; - Tell how electromagnetic principles are applied in energy conversion; - Describe the functioning of momentum of inertia, pulley & belt and gearboxes. - Calculate the steady state operation of system containing mechanical components. - Explain the construction and operation principle of a DC machine; - Explain the nameplate of a given machine; - Apply steady state performance calculation of a DC machine - Explain the common operation principle of AC machines - Explain the construction and operation principle of an induction machine; - Apply steady state performance calculation of an induction machine; - Explain the construction and operation principle of a synchronous machine; - Apply steady state performance calculation of a synchronous machine; - Tell the conditions to connect a synchronous machine to the AC power grid; - Tell the roles of AC machines in modern power grid. - Explain how to use power semiconductors and passive components to convert power; - Tell main DC-DC converter topology (buck, boost) - Analyze DC-DC converters at steady state; - Describe the operation principles of H-brdige inverter and rectifier Power system part: - Describe a power system structure - Model a power system - Describe operation principles during normal conditions - Describe operation principles during disturbances - Apply methods to analyze a power system	
Education Method	32 hrs lectures 2 hrs practical self-grade assignments each week	
Literature and Study Materials	Pieter Schavemaker, Lou van der Sluis, "Electrical Power System Essentials", John Wiley and sons; James L. Kirtley, "Electric Power Principles", John Wiley and sons. Handouts (copies of the lecture slides). Supplementary reading material offered via Brightspace.	
Assessment	1 final exam; 1 lab report, pass/no-pass; A resit. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	Final exam: 3 hrs Resit: 3 hrs	
Permitted Materials during Tests	During the exam, students can use a simple non-programmable and non-graphical calculator (like the TI-30 or Casio FX-82). The exam is NOT an open book exam. During the exam, students are allowed to ask questions to clarify the wording.	
Co-Instructor	Dr. M. Cvetkovic	
Co-Instructor	Dr. J. Dong	

EE1G1	Introduction to Electrical Engineering	5
Responsible Instructor	Dr. I. Ercan	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	This course provides an introduction to Electrical Engineering (EE) and the EE BSc programme at TU Delft. The course is composed of four main components: * lectures (in person) * programming sessions (online) * integrated math (in person) * mentorship (in person)	
Study Goals	At the end of this course, students should be able to: * Define EE fields * Recognize critical challenges in the field * Identify key contributions of EE to society * Classify programming languages * Constructing python codes * Discuss and reflect on teamwork, ethics, academic integrity, study skills (math, time management) * And gain the following mathematical competences / skills: -- Solve systems of linear equations using row reduction; -- Evaluate a one- or two-sided limit of a function at a point or at infinity; -- Interpret the continuity of a function; -- Evaluate derivatives of elementary functions and sums, products, quotients and compositions of these functions; -- Use the derivative to describe the qualitative behaviour of a function; -- Find the linearization of a function; -- Evaluate definite and indefinite integrals of elementary functions and of (piecewise) continuous functions; -- Evaluate definite and indefinite integrals using integration by parts; -- Perform arithmetics involving complex numbers; -- Determine the polar and exponential form of a complex number; -- Find complex solutions of polynomial expressions; -- Interpret a direction field of a first order differential equation; -- Solve a first or second order, linear, homogeneous or nonhomogeneous differential equation or initial-value problem with constant coefficients; -- Describe the movement of a mass-spring system using a second order, linear differential equation infer correspondences with circuit theory.	
Education Method	This course has blended design and is composed of in person lectures, mentorship hours, online sessions. Students are also required to complete take-home assignments.	
Literature and Study Materials	The course does not have a required text book. All necessary reading material will be provided by the instructors during the course.	
Books	Calculus: Early Transcendentals', Interational Metric Edition, 9th edition, by James Stewart, Daniel Clegg and Saleem Watson. ISBN: 9780357113516	
Assessment	This course is a Pass/Fail course. In order to pass the course, students are required to participate in all four components of the course (integrated math, programming, lectures, mentorship) and * Complete all integrated math assignments * Complete all python assignments * Complete all Inspirational Lecture assignments * Attend 75% Mentorship activities and complete 75% of Mentorship assignments In case of insufficient results for the assignments, a repair option may exist in accordance in with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr. F. Sebastiano	
Co-Instructor	Dr. A. Endo	
Co-Instructor	B. Abdikivanani	
Co-Instructor	N. de Kleijn	
Co-Instructor	Dr. W.M. Schouten-Straatman	

EE1L1	Integrated Project 1	5

Responsible Instructor: S. Izadkhast

Contact Hours / Week x/x/x/x: 0/8/0/0

Education Period: 2

Start Education: 2

Exam Period: 2

Course Language: English

Course Contents

In the IP-1 project you and your team will design a complete audio system with "Booming Bass" extension. First, you will measure the characteristics of a 3-way loudspeaker system. You will analyze and build a power supply and a power amplifier. You will design speaker filters and the extension for a so-called "Booming Bass."

Starting from a set of specifications, you will make design choices for each part of the system. Within your team you will discuss the choices and divide the work amongst each other. To verify the choices you will make analytical calculations and numerical simulations. For this purpose you will use the mathematics and the linear circuits theory from previous and concurrent lecture courses.

You will construct the designs and test them with electrical and acoustic equipment. The measurement results are compared with the analytical and simulation results. On this basis, you will conclude to which extent the specifications have been met.

During IP-1 you will also train team working skills. Those include working in organized meetings, making a project plan, giving oral presentations, and delivering technical reports.

Your project results will be graded on the basis of the reports, an oral group presentation, and an individual oral examination.

The sound of the system will be assessed in an informal competition where all students are welcome. A jury of experts will select a winning team and award a small prize.

Study Goals

After completion of this project, you should master the following:

1. Apply basic concepts and models for electrical circuits.
  -(a) Recognize the models of passive elements such as resistors, capacitors, inductors, diodes and an active element namely an operational amplifier,
  -(b) Express the transfer functions of simple electrical circuits of the above mentioned components in the time domain as well as in the frequency domain,
  -(c) Simulation of the circuits in LTspice.

2. Analyse and Build the following basic circuits:
  -(a) First and second order passive electrical filters,
  -(b) Simple opamp voltage amplifier,
  -(c) Power supply with a transformer, diode bridge, capacitors and resistive load.

3. Perform basic practical skills of electronic experimentation:
  -(a) Soldering of components on a printed circuit board based on a schematic,
  -(b) Effective use of a multimeter, function generator, power supply and oscilloscope.

4. Be able to conduct basic experiments:
  -(a) Understanding a specific basic electro technical problem,
  -(b) Retrieving of technical information in data sheets and manuals,
  -(c) Interpretation of measurement results in the light of non-ideal properties of measurement equipment and components.

5. Apply technical communication skills:
  -(a) Give a scientific presentation,
  -(b) Writing of engineering design reports.

6. Apply collaborative problem solving.

Education Method

Practical: group of about 10-12 students with a tutor and a student assistant.
Contact hours (on-campus): 48 hours (6 hours/week) on average.
Self study: 92 hours

Prerequisites: The courselabs of EE1G1 Introduction to Electrical Engineering

Assessment

The IP-1 project will be graded on the basis of the following components:

1. Final report. Group grade; it will constitute 25% of the total grade.
2. Intermediate report. Group grade; it will constitute 15% of the total grade.
3. Individual oral examination in Week 2.9. Individual grade, constituting 30% of the total grade.
4. Oral presentation skills in Week 2.9. Individual grade, constituting 15% of the total grade.
5. Tutor individual grade taking into account your overall performance in the project 15%;
6. Individual tasks during the project. Individual grade, 'Pass' or 'Fail'. During the project all members should have brought to a good end at least the following tasks:
- Giving a presentation on the content of an intermediate report.
- Answering questions on the content of an intermediate report.
7. Mentoring assignment. Individual grade, 'Pass' or 'Fail'. Note that you must also complete information literacy 1 course.

Required for obtaining a final grade for this course is that all the components individually have scored a 'Pass' or a grade of at least

Grading during the project will be done by the tutor. The examinations at the end will be graded by a panel consisting of two or three people who are members of the academic staff of the EE program.

Assessment will be done on the basis of the study goals. You are advised to consult them from time to time in order to structure your study activities.

The deadlines in this project are hard, especially the one for the final report. Work that is handed in too late may still be graded. However, it will be subject to a penalty in such a way that the maximum obtainable grade will be a 6 ('sufficient').

Components that at the first assessment are judged as insufficient, will not be graded. To correct this, one chance will be given, after which a grade will be determined.

IP-1 includes the mentorship. The grade of the IP-1 project becomes not valid until all of the following mentorship requirements have been fulfilled:
- presence and active participation in the mentorate activities.

	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Co-Instructor	Dr.ing. I.E. Lager
Co-Instructor	Ir. E.W. Bol
Co-Instructor	Dr.ir. M. Alonso del Pino
Co-Instructor	S. Izadkhast
Co-Instructor	Dr.ir. R. Vismara
Co-Instructor	B. Abdikivanani
Co-Instructor	Dr. I. Ercan

EE1L2	**Integrated Project 2**	**5**
Responsible Instructor	Dr.ir. A.J. van Genderen	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	EE1D1, EE1D2. The contents of the course EE1E1, that runs in parallel, is also heavily used.	
Course Contents	Build, program and operate a functional autonomous mobile robot. The students will have to apply the acquired knowledge and skills in the area of digital control electronics, sensor technology energy conversion. The IP-2 project 'Smart Robot Challenge' is dedicated to design and build a wireless controlled robot that can autonomously perform a predefined set of tasks. The basic hardware is available at the beginning of the course, i.e. robot platform, Zigbee modules, battery and optical and acoustic sensors. The hardware must be extended such that the robot can transfer electrical energy wirelessly, and the robot must be programmed such that it is able to drive through a maze guided by a programme that runs on a PC. With this, SystemVerilog programming and C programming are employed so that the robot can complete the various challenges and win against other teams robot the final competition.	
Study Goals	Upon successful completion of IP-2 project, the student will be able to: 1. Apply a project-based approach to design an electronic system in a project team; 2. Methodologically design a technical solution to a relevant problem within given boundary conditions; 3. Integrate and apply the knowledge and skills acquired during the lecture classes of the first year, in a design project; 4. Actively apply academic skills in order to intentify and acquire missing information and knowledge, systematically analyze problems, motivate choices, critically interpret results; 5. Document own project-related work and write a well-organized technical report.	
Education Method	Practical: group of 8-10 students with 1 tutor and 2 student assistants. Contact hours: 64 hours (8 hours/week) Self study: 64 hours (8 hours/week + preparation for Final presentation)	
Assessment	1) Final project report 2) Oral group presentation 3) Robot tests/competition In case of an insufficient final result, repair options may exist in accordance with Article 2 Exameneisen, Clause 4, of the Uitvoeringsregeling 2023-2024. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	S. Izadkhast	

EE1M1	Calculus	5
Responsible Instructor	Dr. W.M. Schouten-Straatman	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	Calculus and linear algebra Electricity and magnetism	
Expected prior knowledge	Strongly advised: Introdution to Electrical Engineering	
Course Contents	"In this course the basic concepts from calculus are treated. The course is roughly divided into three parts. The first part of the course focusses on single variable calculus. This parts main subjects are general understanding of functions of a single variable and integration of these functions. General understanding of functions contains a variety of topics, including inverse (trigonometric) functions, limits, derivatives, implicit differentiation, Taylor polynomials and lHospitals rule. Subsequently, the subject of integration mainly focusses on the common integration techniques the substitution rule, integration by parts and integration by partial fraction definition and ends with a treatise of improper integrals. The second part of the course focusses on multivariate calculus. After starting with some general geometrical concepts such as the dot and cross product, as well as graphs and contour plots for functions of several variables, one of the main subjects of this part is partial derivatives and various related topics, including tangent planes, linear approximations, the chain rule, directional derivatives, the gradient and extreme values. This part finishes with a treatise of double and triple integrals, including polar, cylindrical and spherical coordinates, as well as general changes of coordinates. The final part of the course focusses on vector calculus. In this part, vector fields are introduced. In addition, line integrals are covered both for scalar functions and for vector fields and connected to double integrals by means of Greens theorem. "	
Study Goals	"At the end of this course, you should be able to 1. Apply the definitions, theorems and formulas related to the topics mentioned in the Course Contents in concrete mathematical problems. 2. Relate applications within the Electrical Engineering program to these definitions, theorems and formulas. 3. Decompose mathematical problems in relevant subproblems. "	
Education Method	Colstruction (lectures combined with instruction) Typical routine: - Preparation: Students do homework, study a pre-lecture video (or do pre-lecture exercises) - Participation: Students come to the lecture which is a block of two hours. In the lecture a new topic will be introduced by the teacher. During the lecture there will be time to do exercises. Sometimes a practical application of the theory is discussed. - Post-lecture work: Students do homework assignments. These contain book exercises as well as exercises in a digital environment.	
Literature and Study Materials	"Calculus: Early Transcendentals', Interational Metric Edition, 9th edition, by James Stewart, Daniel Clegg and Saleem Watson. ISBN: 9780357113516 Open Linear Algebra (distributed via BrightSpace)"	
Assessment	Examination method: this course will be assessed using examination software with a written component. The exams will contain a mix of short-answer exercises, where only the final answer is graded, and open questions, where the full reasoning and calculation are considered. The examination software (Grasple) will be used; therefore, the exams will take place in a computer room at the TU Delft. The written component will be on paper. Course grade The final grade of the course consists of the following components: - Partial exam 1 This exam covers the material of the first half of the course. This exam lasts 2 hours. - Partial exam 2 - This exam mainly covers the material of the second half of the course. However, some techniques from the first half of the course might be used as well. This exam lasts 2 hours. Exam grade calculation: (0.5 * Partial exam 1 + 0.5 * Partial exam 2) Bonus calculation: At the end of each lecture, a set of practice exercises will be posted on the Brightspace page using the online platform Grasple. Completing each exercise set results in a red, yellow or green check mark, depending on the score. You are allowed to take multiple attempts at each set to improve your score, but the questions may vary slightly per attempt. Participation is voluntary. The bonus grade is based on the Grasple exercises in the following way: each red checkmark counts as a 3, each yellow checkmark as a 6 and each green checkmark as a 10, while a set not made results in a grade equal to 0. For this, the results on the day of the final exam are the ones that are taken into account. The bonus grade is the average over the best 75% of all these grades. Final grade calculation: (exam grade) + ((bonus grade)*(10 - exam grade)/100) Repair possibilities: - Endterm: The resit is taken in one part (i.e., contains all course topics). The result of the sub-exams expire before the resit, and are not transferable to the next academic year. This covers the entire course contents. This exam lasts 3 hours. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Drs. A.T. Hensbergen	

EE1M2	Calculus and Linear Algebra	5

Responsible Instructor	Dr. W.M. Schouten-Straatman
Contact Hours / Week x/x/x/x	0/0/6/0
Education Period	3
Start Education	3
Exam Period	3 5
Course Language	English
Required for	Linear algebra and differential equations
Expected prior knowledge	Required: Calculus
Course Contents	"The goal of this course is to lay a strong foundation for the various and numerous applications of calculus and linear algebra, both directly within the field of electrical engineering as well as in general engineering mathematics. This course consists of two halves: calculus and linear algebra. The first half of this course starts directly where the previous calculus course ended, by introducing parametric surfaces and surface integrals, as well as the curl and divergence of a vector field. Then, all topics in multivariate and vector calculus are connected by means of Stokes theorem and the divergence theorem. Afterwards, the focussed is shifted towards series. These are introduced and analysed by means of various convergence tests, such as the integral test, the alternating series test, comparison tests and the ratio test. Finally, power series are introduced, which results in a treatise of Taylor series. The second half of this course starts from scratch with methods from Linear Algebra to solve and analyse systems of linear equations. Here concepts like vectors, linear combination, span and linear independence play a central role. In addition, matrices and matrix algebra are key concepts that will be discussed, with a major focus on invertibility of matrices. Finally, a lot of time will be devoted towards the more geometrical aspects of linear algebra, such as linear transformations, LU factorisation, subspaces, coordinates, dimension, orthogonality, orthogonal projections and least-squares problems. "
Study Goals	"At the end of this course, you should be able to 1. Apply the definitions, theorems and formulas related to the topics in calculus mentioned in the Course Contents in concrete mathematical problems. 2. Perform calculations in the field of linear algebra. 3. Determine whether a given statement in linear algebra is true or false. 4. Connect various properties of matrices. 5. Relate applications within the Electrical Engineering program to the definitions, theorems and formulas as formulated in the Course Contents. 6. Decompose mathematical problems in relevant subproblems "
Education Method	Colstruction (lectures combined with instruction) Typical routine: - Preparation: Students do homework, study a pre-lecture video (or do pre-lecture exercises) - Participation: Students come to the lecture which is a block of two hours. In the lecture a new topic will be introduced by the teacher. During the lecture there will be time to do exercises. Sometimes a practical application of the theory is discussed. - Post-lecture work: Students do homework assignments. These contain book exercises as well as exercises in a digital environment.
Literature and Study Materials	Calculus: Early Transcendentals', Interational Metric Edition, 9th edition, by James Stewart, Daniel Clegg and Saleem Watson. ISBN: 9780357113516 Open Linear Algebra (distributed via BrightSpace)
Assessment	Examination method: this course will be assessed using examination software with a written component. The exams will contain a mix of short-answer exercises, where only the final answer is graded, and open questions, where the full reasoning and calculation are considered. The examination software (Grasple) will be used; therefore, the exams will take place in a computer room at the TU Delft. The written component will be on paper. Course grade The final grade of the course consists of the following components: - Partial exam 1 This exam covers the material of the first half of the course. This exam lasts 2 hours. - Partial exam 2 - This exam covers the material of the second half of the course. This exam lasts 2 hours. Exam grade calculation: (0.5 * Partial exam 1 + 0.5 * Partial exam 2) Bonus calculation: At the end of each lecture, a set of practice exercises will be posted on the Brightspace page using the online platform Grasple. Completing each exercise set results in a red, yellow or green check mark, depending on the score. You are allowed to take multiple attempts at each set to improve your score, but the questions may vary slightly per attempt. Participation is voluntary. The bonus grade is based on the Grasple exercises in the following way: each red checkmark counts as a 3, each yellow checkmark as a 6 and each green checkmark as a 10, while a set not made results in a grade equal to 0. For this, the results on the day of the final exam are the ones that are taken into account. The bonus grade is the average over the 75% best grades of the Calculus part and the 75% best grades of the Linear Algebra part. Final grade calculation: (exam grade) + ((bonus grade)*(10 - exam grade)/100) Repair possibilities: - Endterm: The resit is taken in one part (i.e., contains all course topics). The result of the sub-exams expire before the resit, and are not transferable to the next academic year. This covers the entire course contents. This exam lasts 3 hours. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Co-Instructor	Drs. A.T. Hensbergen

EE1M3	Linear Algebra and Differential Equations	5
Responsible Instructor	Dr. B.J. Meulenbroek	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	EE1M2 Calculus and linear Algebra	
Summary	Part 1: Linear Algebra: -- Determinants; -- Eigenvalues/Eigenvectors; Diagonalizability of matrices; -- Quadratic Forms; -- Singular Value Decomposition Differential Equations: -- Homogeneous Linear Systems Part 2: Differential Equations: -- Linear Systems: --- Phase Plane --- Matrix Exponential --- Inhomogenous Systems -- Second order linear DEs; Variation of Parameters -- Non-linear Autonomous Systems: --- Phase Plane; --- Trajectories; --- Stability; --- Linearization; Classification of critical points -- Partial Differential Equations: --- Separation of Variables --- Fourier Series --- Heat Equation --- Wave Equation --- Laplace Equation	
Course Contents	Linear Algebra: -Determinants -Eigenvalues and eigenvectors -Diagonalizability -Symmetric matrices -Singular value decomposition Differential equations -First and second order linear differential equations -Systems of linear differential equations -Systems of nonlinear differential equations -Boundary value problems -Fourier series -Partial differential equations, including heat equation, wave equation, Laplace's equation, Maxwell's equations	
Study Goals	At the end of this course, you should be able to: 1. Apply the definitions, theorems and formulas related to Linear Algebra (see course description) in concrete mathematical problems. 2. Relate applications within the Electrical Engineering program to these definitions, theorems and formulas. 3. Decompose mathematical problems in relevant subproblems.	
Study Goals continuation	See Summary	
Education Method	Colstruction (lectures combined with instruction)	
Literature and Study Materials	Required: - Open linear algebra (linked via Brightspace) - Elementary Differential Equations and Boundary Value Problems, by William E. Boyce (any edition from the 9th is suitable)	
Assessment	The final grade of the course consists of the following components: - Written Partial Exam 1 (50%) - Written Partial Exam 2 (50%) Final grade calculation: Final grade = 0.5 * Written Partial Exam 1 + 0.5 * Written Partial Exam 2 Duration of partial exams: 2 hours Repair possibility: Written Final Exam (=Resit) Duration of resit: 3 hours	
Permitted Materials during Tests	None	
Test and result scale	???	
Pop-up tekst bij intekenen	???	
Co-Instructor	Dr. W.M. Schouten-Straatman	

EE1P1	Electricity and Magnetism	5
Responsible Instructor	Dr.ing. I.E. Lager	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	N.A.	
Expected prior knowledge	N.A.	
Course Contents	This course examines the basics of the electromagnetic field in its static and stationary (slowly-varying) states. The course starts by summarising some fundamental prerequisites by insisting on their specific role and impact in the description of electromagnetic field phenomena. The electrostatic interaction is defined by starting from the electric charge (fundamental electric property of matter), the Coulomb force and electrostatic field. It then introduces the ensemble field description via Gauss's law. The energetic description of the electrostatic field is firstly given in terms of the electric potential, then the electrostatic energy (density) is defined and, finally, the capacitors (as energy storage devices) are examined. The description of the electric state is rounded off by introducing the electric current, in its classical interpretation. The second half of the course concerns the magnetic interaction that is firstly evidenced via the pertaining forces and the manners in which the field is produced. The effect of the presence of matter is briefly touched upon. The fundamental laws governing the magnetic interaction, namely Gausss law, Ampères law and Faradays law are discussed at length, together with some pertaining practical applications. The course ends by highlighting the need for an integrated electromagnetic field description in terms of Maxwell's equations.	
Study Goals	After following this course the student should be able to: Estimate, examine and evaluate the basic electric/magnetic field sources: electric charge and electric current. Estimate, examine and evaluate the basic electric/magnetic field quantities: electric field and magnetic field; discriminate their intrinsically VECTOR substance. Estimate, examine and evaluate derived quantities, namely electrostatic/electric potential and the electric and magnetic fluxes; discriminate their intrinsically SCALAR substance. Identify, interpret and examine relatively simple electrostatic configurations; identify sources and calculate forces and potentials by applying superposition. Express Gausss law, and apply it for calculating the electrostatic field in configurations with high degree of symmetry. Examine and quantify the electrostatic energy stored in relatively simple electrostatic configurations; in particular, apply this to capacitors having canonical shapes (primarily, but not restricted to parallel-plate and cylindrical capacitors). Estimate, examine and evaluate the constant electric currents in wire-like and massive conductors; express and evaluate electric currents with variable current densities and/or in nonhomogeneous, massive conductors. Interpret, examine and evaluate magnetic core guided magnetic field systems by organising them into lumped parameter elements making use of magnetic circuit analogies. Identify, interpret and examine the relation between electric field and the current density, and the relation between the current density and the electric charge + drift velocity; discriminate the current densitys intrinsically VECTOR substance. Identify, interpret and examine the magnetic field produced by relatively simple current-carrying, wire-like conductors. Express Ampères law and Faradays law, and apply them for calculating the magnetic field and the induced electric field in configurations with high degree of symmetry. Understand the significance of Gausss law for the magnetic field and infer the inexistence of magnetic monopoles.	
Education Method	32 hrs lectures 16 hrs seminars 6 graded homework assignments (GHAs) 2 partial exams	
Literature and Study Materials	Richard Wolfson: Essential University Physics, Pearson, 4th edition, 2021. Handouts (copies of the lecture slides) are made available after the lectures the lectures are made available. Supplementary reading offered via Brightspace.	
Assessment	This course has: graded homework assignments (GHAs), handled via a TU Delft platform → the GHAs yield a sliding-weight bonus for the exam (details will be communicated at the start of the course) 2 examinations, (weeks 3.5 and 3.10) A resit. The result of the GHAs is expressed as B, and ranges from 0 to 10. R is calculated as the average of the results of all but one of the GHAs, in which any non-made GHA is graded with a 0; either the least-scoring GHA (when all GHAs were done) or a non-made one are omitted from the average. The results of both partial exams are averaged, this gives the exam grade E. The final course grade C is calculated as follows: C = E + B * (10-E) / 100. To pass the course, a final grade C of at least 5.76 points in scale 0-10 is needed. The final grade will be rounded to half points. The GHAs result is valid for the partial exams and the resit, but only in the year the assignments have been made. The resit includes the contents of both partial exams. The result of the partial exams expires for the resit and is not transferable to the next academic year. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	Partial exams: 2 hrs Resit: 3 hrs	
Permitted Materials during Tests	During the exam, students can use a simple non-programmable and non-graphical calculator (like the TI-30 or Casio FX-82). Students are provided a complete list of formulas. The exam is NOT an open book exam. During the exam, students are allowed to ask questions to clarify the wording.	
Co-Instructor	Dr. J. Dong	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bachelor Electrical Engineering**

B-EE jaar 2 2024

Director of Studies	Dr.ir. R.C. Hendriks
Director of Studies	Dr.ir. N.P. van der Meijs
Program Coordinator	Dr. B. Hunyadi
Program Coordinator	C. Thepass
Prerequisites	Let op! De Ingangseisen zijn in de OER vastgelegd. Mocht er verschil zijn tussen wat er hier staat en wat in de OER staat, dan is datgene wat in de OER staat geldig. In het tweede jaar gelden voor de volgende twee studieonderdelen ingangseisen: Je mag pas aan het onderdeel EE2L1 Integrated Project 3 meedoen, als je EE1L1 Integrated Project 1 en het course lab van EE2S1 Signals and Systems, en ten minste EE1C2 Linear Circuits B of EE1M1 Calculus hebt behaald. Je mag pas aan het onderdeel EE2G1 Electrical Engineering for the Next Generation meedoen, als je een positief bsa-advies en de vakken EE1E1 Electrical Energy Fundamentals, EE1M3 Linear Algebra and Differential Equations, EE1L2 Integrated Project 2 en EE2L1 Integrated Project 3 hebt behaald.
Introduction 1	Inschrijven Voor de (deel-)tentamens van álle vakken moet je je uiterlijk 14 dagen vóór de dag van het tentamen via OSIRIS inschrijven. Tussen de 13 en 6 dagen voor de tentamendag is het mogelijk om een late inschrijving te doen. Je komt dan terecht op de wachtlijst en kan deelnemen aan het tentamen als daar plek voor is. Wil je bij nader inzien toch niet meedoen aan een studieonderdeel of tentamen waar je je voor hebt ingeschreven? Schrijf je dan ook weer uit! Zie voor uitleg https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens
Introduction 2	Vooruitblik jaar 3: In het derde jaar volg je tijdens het eerste semester een minor. Deze kies je al dit jaar. Kijk voor meer informatie op http://minors.tudelft.nl. Wil je in je derde jaar naar het buitenland? Begin je dan nu al te oriënteren op https://www.tudelft.nl/studenten/ewi-studentenportal/onderwijs/study-abroad Zie ook de Brightspace van Study Abroad EEMCS

EE2C1	Transistor Circuits	5
Responsible Instructor	Dr. S. Du	
Responsible Instructor	Dr.ir. M.A.P. Pertijs	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course extends the analysis of electronic circuits to circuits containing non-linear components as key building blocks for realizing essential functions such a switching and amplification. Noise is introduced as a fundamental limitation to signal quality, motivating the need for signal amplification. The physics of diodes and MOSFETs is explained at an intuitive level, and their characteristic equations are introduced. Large-signal operation is presented as approach to realize digital logic, with the CMOS inverter as basic building block. Biasing and small-signal operation are presented as approaches to obtain linearised behavior from non-linear circuits, and are applied to single-MOSFETs amplifiers (common-source, common-gate, common-drain stages). Amplifiers with feedback are introduced as a means of realizing ideal transfer functions. Finally, opamp-based amplifiers and single-MOSFET amplifiers are shown to be practical approximation of these ideal amplifiers.	
Study Goals	After following this course, you should be able to: 1. Calculate noise levels and SNR based on the thermal noise of resistors. 2. Describe the behaviour of diodes and MOSFETs based on their characteristic equations. 3. Analyse the static large-signal behaviour of simple non-linear circuits, including the CMOS inverter. 4. Analyse non-linear circuits using small-signal analysis. 5. Calculate the small-signal parameters of single-transistor amplifiers based on MOSFETs. 6. Synthesize ideal amplifier transfer functions using passive feedback around active amplifiers. 7. Synthesize CMOS logic circuits to implement Boolean operations	
Education Method	Lectures, instructions, practical (course lab), Self-graded homework	
Literature and Study Materials	"Microelectronic Circuits" by Sedra, Smith, Carusone and Gaudet. 8th international edition, ISBN 9780190853501	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) - Lab assignments (graded pass/fail) Repair possibilities: - A resit - Resit for the course labs In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	Final: 3 hours Resit: 3 hours	
Co-Instructor	Dr.ir. M.A.P. Pertijs	
Co-Instructor	Dr. S. Du	

EE2C2	Mixed-Signal Circuits and Systems	5
Responsible Instructor	M. Babaie	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Building on the concepts taught in Transistor Circuits, this course introduces a number of key building blocks of mixed-signal systems. The course starts with examples of mixed-signal systems (e.g. electric vehicle, smartphone, biomedical implant), highlighting the role and importance of these building blocks. It introduces power conversion based on linear- and switched-mode regulators and switched-inductor (buck/boost) converters. The CMOS inverter is extended to CMOS implementation of combinatioral and sequential logic, and analysis of the associated timing and performance limiting factors. Small-signal modeling of the MOSFET is extended to include high-frequency behavior in order to analyse the frequency response of single-stage amplifiers. Analog-to-digital and digital-to-analog converters are introduced as interfaces between analog and digital circuits, including the basic operating principles of SAR and dual-slope ADCs. Finally, several commonly-used oscillators and filters are introduced.	
Study Goals	After following this course the student should be able to: 1. Describe the role for key building blocks of mixed-signal systems in an application context. 2. Analyze power conversion circuits, including linear regulators, buck converters and boost converters. 3. Analyze CMOS combinational and sequential logic circuits. 4. Calculate small- and large-signal behaviour of amplifiers using high-frequency transistor models. 5. Understand the operation of analog-to-digital converters (ADCs) and digital-to-analog converters (DACs). 6. Calculate filters transfer function and oscillators operating frequency. 7. Analyze sensor readout circuits.	
Education Method	Lectures, instructions, courselab, Self-graded homework	
Literature and Study Materials	1) Microelectronic Circuits, SEVENTH EDITION, Adel S. Sedra, 2) A. Johns Kenneth W. Martin (ADC/DAC) 3) DCDC Converters - POWER ELECTRONICS, Dariusz Czarkowski For further reading: Analog Integrated Circuit Design, Tony Chan Carusone David	
Assessment	The final grade of the course conssits of the following components: Written final exam (100%) A written lab report for course labs is graded as a pass or fail. Repair possibilities: A resit for the exam Corrections for the lab report In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	Final exam: 3 hours Resit: 3 hours	
Co-Instructor	Dr. S.M. Alavi	
Co-Instructor	C. Gao	

EE2G1	Electrical Engineering for the Next Generation	10
Responsible Instructor	Prof.dr.ir. L. Abelmann	
Contact Hours / Week x/x/x/x	0/0/0/12	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	EE1L1 Integrated Project 1, EE1L2 Integrated Project 2, EE2L1 Integrated Projet 3, en EE1E1 Electrical Energy Fundamentals, en EE1M3 Linear Algebra and Differential Equations	
Course Contents	In the Electrical Engineering for the Next Generation project, you have the last chance to prepare yourself well for the bachelor end project. You have already had training in systematic design, teamwork, societal and ethical aspects, on which we will expand. There is one more important skill you will need: learn how to learn. You will expand your skills and knowledge in one of four exciting sub-projects, addressing different societal challenges and a variety of new knowledge to be acquired. Precision Agriculture Within this project, you will realize a soil monitoring system for precision agriculture. The project involves: (i) The development of mobile autonomous sensors mounted in a rover. (ii) The development of a communication system to collect sensor information in space and time. (iii) Signal and data processing to compare the sensor's information with reference sensors and correlate that information spatially and temporally. You will study and specialize yourself in one of the sub-systems, such as sensors analog front-end, data acquisition, communication standards/protocols, signal processing techniques. Communication and sensing By successfully completing this project, you will be able to analyze RF up/down-converting circuitry, to model and design antennas, and familiarize with the algorithms employed in modern radar and communication systems. You will apply theoretical concepts of electromagnetic and wireless systems for communication & sensing, and you will design, assemble, and characterize components and building blocks employed in communication & sensing systems. Health Within this project you will realize a human machine interface to a wheelchair. The project involves signal acquisition (EEG helmet, EMG pads), signal processing and mechatronics (wheelchair). You will study and specialize yourself in one of the sub-systems, such as biosignals, low noise amplifiers, machine learning, bus communication standards and mechantronic control. Sustainable Energy By successfully completing this project, you will be able to design and size a photovoltaic solar farm, select the suitable power converter topology to condition the DC-generated power from the solar farm for integration into the electric system. You will also design and rate the solar farm cabling system and acquire knowledge about the components used in high-voltage substations. Moreover, you will learn the decision-making process for the seamless integration of energy into a grid with a certain set of constraints. You will specialize yourself in one of the sub-topics, i.e. photovoltaic energy generation, power conversion, high voltage transmission and distribution grid, and power system analysis. Detailed information about the learning activities in the four sub-projects can be found on Brightspace.	
Study Goals	After following this course you should be able to: LO1. Acquire a new field of knowledge relevant to EE and your project independently. LO2. Design a solution with several sub-systems for a complex problem using the systematic design methodology. LO3. Apply teamwork techniques that allow you to form effective teams that communicate actively and constructively and enable collaborative learning. LO4. Formulate design requirements from an ethical perspective while considering multiple stakeholders.	
Education Method	Project, Lab, Instructions, PBL, Lectures	
Assessment	The assessment of the knowledge part will be a based on an individual oral assessment and a written report, and counts for 50% for the final grade. Repair option: oral examination The evaluation of the design and teamwork learning objectives (LO2-4) will be based on reports and presentations, accounting for the remaining 50% of the final grade. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	M. Spirito	
Co-Instructor	F. Arroyo Cardoso	
Co-Instructor	S.A. Rivera Iunnissi	

EE2L1	Integrated Project 3	5
Responsible Instructor	Prof.dr.ir. A.J. van der Veen	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	Linear algebra, signals & systems. Prerequisites: EE1L1, the course lab of EE2S1, and, at least EE1C2 or EE1M1.	
Course Contents	The lab consists of two parts: 1. Two Python computer exercises, which continue the course labs of EE2S1 Signals and Systems: - recording dual-microphone audio signals - deconvolution (matched filter, frequency domain approach) - application to estimating the distance between two microphones using an audio signal 2. Design of a complex system, with the choice of two projects: A. KITT self-driving car B. Electronic stethoscope A In this project you team with your fellow students to remotely operate an electric toy car called "KITT". The car is equipped with ultrasonic parking sensors, a Bluetooth connection, an audio beacon and a micro-controller. You are in charge of this "enhanced" car and will design and implement algorithms so that the car can localize itself, sense obstacles and find its way autonomously; as well as to communicate with a base station at which calculations are performed. High tracking accuracy and speedy and tight control are your objectives. In a series of competitions you and your team will show your achievements in challenges of increasing difficulty. B In this project you work with a 6-channel digital stethoscope array. The first objective is to capture heart signals, filter and analyse the signals (anomaly detection such as murmurs). The next objective is to localize the sources (e.g., the 4 valves) using deconvolution, TDOA estimation, and MUSIC-like (eigenvalue-based) algorithms. The algorithms are designed and tested on a dummy. After consideration of the ethical aspects, the signals could also be acquired on human subjects. The projects contain a midterm report and are wrapped up by a demo session and a final report, followed by a presentation/discussion in front of a committee.	
Study Goals	After following this course you should be able to: 1. Analyze a real-world application which involves both hardware and software by breaking it down into smaller modules. 2. Apply signal processing (and linear algebra, statistics, modeling, communication, control) tools to design and implement the solution using the python programming language. 3. Design a complex system using design methodology as developed over the various IP labs. 4. Plan, manage and conduct effective teamwork for the successful realization of a technical complex assignment. 5. Write and present a technical report that complies with current standards of the electrical engineering scientific community. 6. Reflect on ethical aspects associated with the application of the designed system in society.	
Education Method	Practical	
Literature and Study Materials	IP-3 Manual	
Assessment	The final grade of the project consists of the following: - Assignments (p/f) - Midterm report (0.3) - Demo/competition (0.2) - Final report, oral presentation and defence (0.5) The final grade is the weighted sum of these components. Individual grades can be adjusted via a peer-review system and staff observations, which can lead to a maximum 2 point variation (-1 to 1) with respect to the average group grade. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	B. Abdikivanani	

EE2M1	Probability and statistics	5
Responsible Instructor	Dr. C.O. Smet	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Calculus, in particular differentiating and integrating functions of one and two variables is indispensable.	
Course Contents	Graphical and numerical summaries of data Introduction to probability Some elementary combinatorics to find probabilities Conditional probability; Bayes' rule; stochastic (in)dependence Random variables; probability mass function; density function; distribution function Standard distributions: Binomial, Poisson, Geometric, Normal, Uniform, Exponential, Pareto Expectation and variance; transformations; Jensen's inequality Multivariate random variables; joint distributions; joint and marginal density functions; (in)dependence of random variables Covariance and correlation Chebychev's inequality; Law of Large numbers; Central Limit Theorem Sampling theory and statistical models; mean, sample variance, histogram, empirical distribution function, boxplot Theory of Estimators: Bias, Efficiency, Mean squared error Linear Regression: univariate and multivariate, goodness of fit Introduction to statistical learning Confidence intervals of Mean and Proportion; t-distribution Testing theory: Type I/II Error, p-value, significance level, critical region One sample t-test Parametric bootstrap	
Study Goals	At the end of the course, the students will be able to: - compute (conditional) probabilities - use discrete and continuous random variables - use joint random variables - compute expectation and (co)variance - apply the law of large numbers and the central limit theorem - analyse data by means of graphical and numerical summaries - formulate a statistical model - estimate parameters - perform a linear regression analysis - construct a confidence interval - formulate and carry out hypothesis tests - perform a bootstrap analysis - Use Python to perform elementary simulation and data analysis	
Education Method	Education will take place in a blended way: the student watches a prelecture video, makes some exercises related to this, participates in the lecture (which consist of lecturing by the teacher and making exercises) and then makes some more exercises at home. This cycle is repeated for every lecture.	
Literature and Study Materials	We use the "Blue Book": A Modern Introduction to Probability and Statistics Understanding Why and How Series: Springer Texts in Statistics Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E. 2005, XVI, 488 p. 120 illus., Hardcover ISBN: 978-1-85233-896-1	
Assessment	The final grade of the course consists of the following components: - Midterm on Probability Theory (50%) - Endterm on Statistics (50%) Final grade calculation: (0.5 * Midterm + 0.5 * Endterm) Repair possibilities: - Resit in next quarter Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Permitted Materials during Tests	Formula sheet (one sheet, to be made by the student) and a simple, non-programmable, non-graphical calculator.	
Co-Instructor	N. de Kleijn	

EE2P1	Electromagnetics	5
Responsible Instructor	Dr.ir. R.F. Remis	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course deals with electromagnetic fields and electromagnetic wave propagation in free space and in materials, reflection and transmission effects at interfaces, induced phenomena (such as heat production, charge accumulation, and surface current generation), transmission lines, voltage and current distributions in transmission lines and radiation by elementary sources. Topics: Review of vector algebra and calculus, electrostatics, magnetostatics, Lorentz's force law. Maxwell's equations in vacuum, time and frequency representation, causality, conservation of charge, time-harmonic electromagnetic fields, polarization state Maxwell's equations in matter, induced currents and the constitutive relations, the electromagnetic boundary conditions Poynting's theorem for transient and time-harmonic waves Uniform plane waves, TE and TM electromagnetic waves, reflection and transmission at planar interfaces, the Fresnel reflection and transmission coefficients, Brewster angle and total reflection The electromagnetic field in a highly conducting material, eddy currents, skin effect, induced surface current and surface impedance Transmission lines and the basic transmission line equations (telegraph equations), transverse electromagnetic waves (TEM waves), characteristic impedance, the coaxial line and the parallel-plate waveguide, propagation along lossless and lossy transmission lines, Radiation and reception, elementary antenna theory, Friis equation	
Study Goals	After following this course you should be able to: 1. Explain the concepts underlying the Maxwell equations and their consequences for electrical engineering practice. 2. Calculate the electric and magnetic fields associated to electromagnetic waves in free space, dielectric/conducting material and transmission lines. 3. Calculate the power transported by electromagnetic waves in free space, dielectric/conducting material and transmission lines. 4. Calculate electric and magnetic fields radiated by elementary antennas in free space. 5. Get acquainted/experience with basic macroscopic electromagnetic phenomena and basic electromagnetic propagation properties of through a series of elementary experiments.	
Education Method	Lectures, Instructions, practical (lab)	
Literature and Study Materials	Ulaby and Ravaioli "Fundamentals of Applied Electromagnetics" Pearson	
Assessment	The final grade of the course consists of the following components: - Written Partial Exam 1 - Written Partial Exam 2 Practical/lab: pass/fail. Final exam grade = 0.5 * Written Partial Exam 1 + 0.5 * Written Partial Exam 2 Exam grade is valid provided a pass is obtained for the lab. Repair possibilities: Written Final Exam In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	O.A. Krasnov	
Co-Instructor	Prof.dr.ing. A. Neto	
Co-Instructor	Dr.ir. M. Alonso del Pino	

EE2P2	Semiconductor Physics and Devices	5
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Course Contents	This lecture course focuses on some of the most important materials and devices used in electrical engineering: semiconductors and semiconductor devices. Semiconductor devices play a pivotal role in electrical engineering: from light-emitting diodes to solar cells, from transistors to computer chips. This lecture course introduces quantum mechanics and solid-state physics in order to understand the material properties of semiconductors and hence to delve into the physics of the operation of semiconductor devices. Elements of quantum mechanics will be presented to explain the wave properties of electrons and the discrete nature of energy levels. This understanding will then be used to develop the solid-state-physics for semiconductors, in particular the band structure of common semiconductors, the density of states, and the concepts underlying the transport of charge carriers in these materials. Attention will be paid also to the manipulation of semiconductor properties by extrinsic doping. Using the essential knowledge on solid-state physics for semiconductors, attention will be focused on semiconductor devices. First the movement of charge carriers in doped semiconductors under the influence of generation, recombination, drift, and diffusion will be discussed. This knowledge will be applied to p-n junctions, which will provide the basis for a discussion on solar cells and light-emitting diodes (LEDs), as well as a brief discussion on the bipolar junction transistor (BJT). In addition, the combination of semiconductors with insulators and metals will be discussed, leading to the metal-oxide-semiconductor field-effect transistor (MOSFET) that forms the basis of the microelectronics industry today.	
Study Goals	After following this course you should be able to: 1. Explain the discrete nature of energy levels in confined systems using concepts of quantum mechanics. 2. Describe energy bands and electronic transport in semiconductors using concepts of solid-state physics. 3. Evaluate function, characteristics and design of basic semiconductor devices using band diagrams and equivalent electric circuits.	
Education Method	Lectures, instructions	
Literature and Study Materials	Donald A. Neamen, Semiconductor Physics and Devices: Basic Principles, fourth edition. M. Mastrangeli and J. Hoekstra, Quantum mechanics reader (made available by the instructors).	
Assessment	The final grade of the course consists of the following components: Non-mandatory assignments and written examination. The final mark for this course, F, will be based on the written examination, E, and the average result of non-mandatory assignments, A. All marks will be expressed on a scale of 0 to 10. For calculating A, first the lowest mark of all non-mandatory assignments will be discarded, and the mark will be determined by calculating the average of the three remaining assignment results. Non-mandatory assignments that are not handed in or not handed in on time will have a score of 0. The results for the non-mandatory assignments only will be valid within the academic year in which these assignments have been made. Results for non-mandatory assignments obtained in previous academic years will not be taken into account. The final mark, F, for this course will be computed as follows: $F = E + [A \times (10 - E)/100]$. The computation will be carried out using one decimal and subsequently rounded to halves. Repair possibilities: - Resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025.	
Co-Instructor	Dr. R. Ishihara	
Co-Instructor	M. Mastrangeli	

EE2S1	Signals and Systems	5
Responsible Instructor	Prof.dr.ir. A.J. van der Veen	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	A Pass on the course labs of EE2S1 is required to participate in the EE2L1 Integrated Project 3.	
Summary	Starting from complex function theory, this course develops the mathematical description of signals and linear time-invariant (LTI) systems by means of the Laplace and Fourier transforms. In this description, signals are represented by sums of complex exponentials, being the 'eigenfuctions' of LTI systems. It follows that the effect of an LTI system on a signal (convolution) can equivalently be described by a product in the Laplace or Fourier domain. The implications are profound and form the basis of a large part of electrical engineering (and other engineering studies). This course is the basis for follow-up courses such as Digital Signal Processing. The course covers the Laplace, Fourier and z-transform, and presents the relations between signals in time domain and frequency domain, first for time-continuous (analog) signals, and then for time-discrete (digital) signals. The course also covers the basics of analog filter design (analog filter functions, IIR filter design, Butterworth and Chebyshev filters), digital filter design via transformation of analog filters to digital filters (impulse invariance, bilinear transform, frequency transformations), and simple digital filter structures. The three course labs in Part 2 cover convolution, frequency domain, and filter design. They use Python programming and Google colab scripts.	
Course Contents	The course presents the theory of signal transformations and linear time-invariant systems, preparing for courses on signal processing, systems theory, and telecommunication. Part 1: continuous-time signals - Linear time-invariant systems (LTI), eigenfunctions and eigenvalues of an LTI system, Dirac delta function, Heaviside unit step function. Convolution. - Laplace transform, region of convergence. Properties. Convoluton and product, inverse transform. - Periodic signals. Fourier series, line spectrum. - Fourier transform (continuous time). - Transformations of a pulse, sinc function, sign function, step function. Duality of the Fourier transform. Shift and modulation, Parseval's theorem. Part 2: discrete-time signals - Ideal sampling. impulse train, periodic spectrum. Sampling theorem (Nyquist, Shannon). Ideal reconstruction - Discrete signals and systems, convolution. - Z-transform - Realizations (direct form for FIR, IIR) - Fourier transform (discrete time) - Filter design (analog and digital), frequency transformations. Part 2 also contains 3 course labs that train you in the use of convolution, frequency domain (Fourier transform) and filter design. The course labs are based on Python programming.	
Study Goals	After following this course you should be able to: 1. Understand and apply integral transformations that play a role in LTI signals and systems. 2. Explain and apply continuous-time convolution and transformations: Laplace transform, Fourier Series, Fourier Transform. 3. Explain and apply the theory of ideal sampling and reconstruction, and the relations between continuous-time and discrete-time spectra. 4. Explain and apply discrete-time convolution and transformations: z-transform, Discrete Time Fourier transform. 5. Design analog and digital filters from frequency-domain specifications 6. Design discrete-time (direct) realizations from a given transfer function.	
Education Method	Lectures, course labs	
Literature and Study Materials	Required: Signals and Systems using MATLAB, Third Edition" by Luis Chaparro and Aydin Akan, Academic Press; 3rd edition (2018). ISBN: 978-0128142042	
Assessment	The final grade of the course consists of the following components: - Written partial exam 1 (50%) - Written partial exam 2 (50%) - Lab assignments (graded pass/fail) Repair possibilities: - A resit - Adjusted deadline for the course labs In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Dr.ir. R.F. Remis	
Co-Instructor	B. Abdikivanani	

EE2S2	Systems and Control	5
Responsible Instructor	Dr.ing. S. Grammatico	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Course Contents	In this course, we study properties of dynamic systems and how these properties can be influenced via automatic control. To this aim, we develop mathematical models for the dynamic behavior of systems, with focus on linear models. Then, we analyze the important system properties both in the time domain and in the frequency domain. Next, we introduce the concept of feedback control and discuss its effects on the system properties. Finally, we present some feedback control approaches, with an emphasis on proportional-integral-derivative controllers.	
Study Goals	After following this course you should be able to: 1. Describe simple physical systems (e.g. electrical, mechanical, electro-mechanicalsystems) via mathematical dynamic models. 2. Analyze the dynamic behavior of closed-loop systems described by state-space models or transfer functions in terms of step and impulse responses, zeros and poles. 3. Interpret Bode diagrams of 1st and 2nd order systems, and sketch Bode diagrams of systems given via a transfer function. 4. Design controllers for linear dynamical systems. 5. Apply the above approaches and methods via dedicated software.	
Education Method	Lectures; Instruction	
Literature and Study Materials	Required: G. F. Franklin, J. D. Powell, and A. Emami-Naeini, Feedback Control of Dynamic Systems 7th edition	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) - Lab assignments (graded pass/fail) Repair possibilities: - A resit - Resit for the course labs In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	G. Pantazis	

EE2T1	Telecommunication and Sensing	5
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Analysis (in particular integration, differentiation and complex calculation). Signals and Systems: LTI-systems, convolution, impulse response, Fourier-transformation and filtering. Probability and Statistics: probability model, conditional probability, independence, probability density function (PDF), expectation and variance, auto-correlation and power spectral density (PSD) function.	
Course Contents	The aim of this course is for you to gain basic insight and understanding of modern telecommunication and sensing systems with a unique combination of physics and signals & systems theory. The course starts with the description of fundamental concepts in telecommunication and sensing including channel capacity, received signal power and noise, link budget, signal modulation and detection, and range-doppler profile. Baseband techniques and bandpass signals with analog and digital modulation methods are discussed in detail. The last part of this course provides an introduction to data communication networking aspects. The lectures are supported by practical numerical problems throughout the course as weekly homework exercises and in the instruction lectures, in order to make you skilled in solving system related problems supported by calculations. The list of topics and distribution over the lectures is as follows: L1. Introduction: a general introduction on modern communication and sensing systems, OSI model, ideal communication and ranging systems, channel capacity (Shannon). L2-3. Wireless and guided links and noise: propagation of e.m. waves, antenna radiation pattern, polarization, received signal power in communication and sensing systems, thermal noise sources, antenna noise, system noise calculation and link budget evaluation L4. Properties of signals: review ideal/practical signals, dB relations, power spectral density, distortionless transmission and bandwidth definitions. L5. Signal sampling and PCM: Signal sampling, natural sampling, flat-top Pule Amplitude Modulation, Pulse Code Modulation: digital encoding, Signal-to-Noise ratio for PCM. L6. Baseband signaling: description of digital information carrying waveforms, line codes and eye pattern, spectral efficiency, inter-symbol interference (ISI). L7. Bandpass signals and systems, distortionless transmission and group delay, bandpass sampling, superheterodyne and homodyne receivers. Analog modulation techniques: Amplitude Modulation (AM), Double Sideband - Suppressed Carrier (DSB -SC) modulation. L8. Analog modulation techniques (continued). Phase Modulation (PM) and Frequency Modulation (FM). Signal-to-Noise Ratio (SNR) after detection for analog modulation techniques. Envelope and product detector. L9. Digital modulation techniques: Binary techniques: OOK, BPSK, DPSK, FSK, Multilevel modulation: QPSK, MSK, QAM. L10. Optimum signal detection for communication and sensing. Optimal receiver, matched filter receiver, waveform ambiguity function. L11. Bit Error Probability framework, BER in baseband systems: NRZ unipolar, polar, bi-polar using a lowpass filter and matched filter. L12. BER for digital bandpass signals using a for coherent (matched-filter) demodulation and non-coherent detection: OOK, BPSK, FSK, DPSK, QPSK, MSK. L13. Fundamentals of sensing: ranging, doppler, range-doppler plane. L14-15 An introduction to the OSI (Open Systems Interconnect) layered model and discuss the OSI layers and their protocols.	
Study Goals	After following this course you are able to: 1. Explain the basic working principles of communication and radar systems, as well as the key similarities and differences between the two applications. 2. Apply the fundamentals of information, signal and EM theory in the performance analysis of simple communication and radar scenarios to estimate quantitatively, (1) signal-to-noise ratio and bit error rate in communication systems, (2) range and velocity of objects in sensing systems. 3. Solve communication and radar system related problems supported by calculations.	
Education Method	Lectures and working lectures. Recorded lectures will be made available via Collegerama. To support studying of the material of this course, exercises are made available for each lecture and each working lecture via Brightspace. These exercises with answers will help you in mastering the course material.	
Literature and Study Materials	Couch, L.W., Digital and Analog Communication Systems, and handout or copies of same chapters of other books, e.g. T.Rappaport, Wireless Communications: Principle and Practice, G. Stimson, Introduction to Airborne Radar, P. Van Mieghem, Data Communications Networking, Chapters 1-7, 2011, ISBN: 978-0-273-77421-1.	
Assessment	The course Telecommunications and Sensing is completed with a 3-hour closed-book written exam covering the full course content at the end of the 2nd or 3rd quarter (resit). The final grade of the course Telecommunications and Sensing is based solely on the grade obtained for the exam, which is is rounded to the nearest half. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	During the 3-hour closed-book exam you are allowed to use the following materials: - a non-programmable electronic calculator, - the overview Tables and Figures to be used at the exam Telecommunications and Sensing (EE2T1), as made available on Brightspace, - an original self-handwritten equation form: 1 page A4, written on both sides, (not type-written, no copy and it should not contain worked out exercises or examples).	
Co-Instructor	Prof.dr. O. Yarovyi	
Co-Instructor	M.A. Kitsak	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bachelor Electrical Engineering**

Electives

EEX01	Introduction to Machine Learning	5
Responsible Instructor	B. Abdikivanani	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course is an elective aimed at 2nd (or 3rd) year BSc EE students. During the course students are exposed to the basics of machine learning (ML) suited for electrical engineering students. The topics include: Mathematical optimisation, regression and classification, linear unsupervised learning, neural networks and tree-based learners. We will use the Python programming language in this course.	
Study Goals	After following this course you should be able to: 1. Explain the theoretical concepts behind machine learning algorithms and its challenges including data, feature extraction, model selection and evaluation. 2. Formulate practical real-world problems as mathematical equations (optimization problems) and solve them. 3. Design and implement various machine learning algorithms for real-world applications using python.	
Education Method	lectures, instructions, weekly assignment, project	
Assessment	The final grade of the course consists of the following components: - project report (0.5) - final exam (0.5) - assignments (p/f) Final grade calculation: (0.5 * project + 0.5 * written exam) * students should receive a minimum of 6 (out of 10) in each item to pass the course In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. J.L. Cremer	

EEX02	Communication Networks and Algorithms	5
Responsible Instructor	M.A. Kitsak	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an introduction to telecommunication networks, which includes local area networks, wireless/mobile networks and the Internet. The course will introduce the OSI and TCP/IP protocol suites and will describe the operation of the OSI layers. During the course, will discuss, among other topics, the operation and concepts of the Data Link, Network, Transport, and Application layers of the OSI model. To address communication network problems effectively, our typical approach begins with crafting a suitable model tailored to the problem at hand. Once the problem is accurately modeled using an appropriate data structure, attention can be directed towards devising solutions for the modeled scenario. The selection of the most suitable data structure holds paramount importance in generating efficient and, importantly, optimal solutions. Typically, solutions for such problems manifest in the form of algorithms. Hence, our journey will encompass techniques for problem modeling, algorithm selection, and the utilization of various data structures such as graphs and trees.	
Study Goals	At the end of the course, you should be able to: 1. Explain the functionalities of modern end-to-end communication networks. 2. Explain the architecture and design concepts that underlie the Internet/communication networks and their protocols. 3. Explain the basic tradeoffs and design/optimization challenges of modern communication networks. 4. Perform elementary calculations and performance analyses on small networks. 5. Implement basic mathematical coding and routing algorithms that have inspired error correction methods and routing protocols in networks. 6. Create models for specific problems. 7. Develop algorithmic solutions for network communication problems and Analyse those algorithms. 8. Employ graph theory (e.g., graphs, tress, and networks) to solve networks communication problems.	
Education Method	Lectures, Discussion meetings, Homework exercises, Team projects	
Literature and Study Materials	Hand out lectures, solved exercises, and additional reading material. Strongly advised: - P. Van Mieghem, Data Communications Networking, Chapters 1-7, 2011, ISBN: 978-94-91075-01-8. - Discrete Mathematics, 8th Ed. - Richard Johnsonbaugh - Pearson (2017).	
Assessment	The final grade of the course consists of the following components: Written final exam (80%), Mini team projects (5% each), homework exercises (10% total) T= (0,8 * Written exam + 0,05 * Mini team project 1 + 0,05 * Mini team Project 2 + 0.1 * Homework exercises) Repair possibilities: - Resit In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Prof.dr.ir. S. Hamdioui	
Co-Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Co-Instructor	Dr. R. Litjens	
Co-Instructor	Dr.ir. E. Smeitink	
Co-Instructor	C. Galuzzi	

EEX03	Microwave Engineering	5
Responsible Instructor	Dr. D. Cavallo	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The course deals with fundamentals of modern RF and microwave engineering. The main topics are transmission lines, microwave networks, impedance matching, introduction to antennas, RF simulation and design techniques.	
Study Goals	You should be able to: 1. Represent microwave networks with matrices (scattering, impedance, admittance, transmission matrices). 2. Formulate electromagnetic waves on transmission lines 3. Solve impedance matching problems and present the solution. 4. Analyze basic passive networks (couplers, phase shifters, power dividers, ...) 5. Solve a problem using RF simulation tools	
Education Method	Lectures, Instructions, Project	
Assessment	The final grade of the course consists of the following components: - Oral exam (80%) - Report for course labs (20%) G= Grade oral exam * 0,8 + Report assignments * 0,2 Repair possibilities: Repair oral exam and report In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Y. Aslan	

EEX04	Technologies for Energy Transition	5
Responsible Instructor	S.A. Rivera Iunnissi	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course will introduce students to system architecture, basic principles, and building blocks of electric power generation from renewable energy sources. Starting with an overview of modern energy systems, it covers various types of power electronics converters that enable the energy transition towards a more sustainable future. Furthermore, students will learn how to estimate the potential of renewable energy sources to meet humanity's energy consumption using the physical principles of energy generation and consumption. After taking the course, students can analyze, model, simulate, and operate power electronics-based energy systems and acquire the fundamentals to design/size them.	
Study Goals	At the end of the course, you should be able to: 1. Understand the principles of energy generation, storage and consumption; 2. Make simple estimations/calculations of the potential of (renewable) energy sources in reference to humankind energy consumption using simple physical tools/concepts. 3. Describe the operation principle and structure of basic building blocks of a power electronics based system; describe hardware and control architecture for regulating basic power converters. 4. Identify the system topologies of power electronics based systems enabling renewable energy transition; 5. Analyse basic power electronics converters at steady state based on flux linkage and charge balance; 6. Calculate voltage/current stress of critical components during normal operation of basic power converters and tell the sizing conditions of components based on it; 7. Carry out simulations and simple experiments of basic power conversion systems, and evaluate the results;	
Education Method	Lectures, practical, Homework assignments	
Literature and Study Materials	Required: Y. Yang, K. A. Kim, F. Blaabjerg, A. Sangwongwanich, Advances in Grid-Connected Photovoltaic Power Conversion Systems, Woodhead Publishing. (9780081023396) Required: David JC MacKay, Sustainable Energy - Without the hot air. (9780954452933) Recommended: J. G. Kassakian et al., Principles of Power Electronics, Cambridge University Press.	
Assessment	The final grade of the course consists of the following components: - Written final exam (80%) - Assignments (20%) T = grade for written final exam * 0,8 + grade for assignments * 0,2 The lab report for course labs is graded as a pass or fail. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Prof.dr.ir. P. Bauer	
Co-Instructor	Prof.dr.ir. A.H.M. Smets	
Co-Instructor	M. Rana	

EEX05	Chip Design	5
Responsible Instructor	Dr.ir. A.J. van Genderen	
Responsible Instructor	Prof.dr.ir. G. Gaydadjiev	
Responsible Instructor	Dr.ir. M.C.R. Fieback	
Responsible Instructor	Dr.ing. C.P. Frenkel	
Responsible Instructor	C. Gao	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Update 2nd December 2024: EE1D1 Digital Systems A and EE1D2 Digital Systems B	
Course Contents	During this project, the student will learn how to make a chip design with a group. The project focuses on structural hierarchical design and on working together within a group. The students are divided into groups of about 5-7 persons and each group together designs a chip. All groups will have to select one or more sensors from a list of three options.	
Study Goals	At the end of the project, you should be able to: 1. Make a design using global product specifications, including boundary conditions, 2. Design in a systematic hierarchical way 3. Use analysis together with synthesis during the design process 4. Analyse their design at different levels of abstraction, 5. Use different types of models during the design, 6. Take into account testability 7. Perform the above design process within a group, 8. Make a design as a part of a group design, 9. Use modern computer tools during the design, 10. Make documentation about the design.	
Education Method	Project (3.1-3.9), introduction lecture, guest lecture on sensors (3.1)	
Literature and Study Materials	Project manual and other information on Brightspace	
Assessment	A grade is given for a module design report that is written halfway during the project, by 2 or 3 students, about a part of the design. A group grade is given for the total group design. This grade is based on the design itself and the report about it, the presentation and the defense. The group grade is adjusted by a peer review that is held. When G is the group grade, P is the possible adjustment made by a peer review (-1 <= P <= 1), and M is the grade for the module report, then the final individual grade is computed as 0.8 x (G + P) + 0.2 x M.5 A group grade is determined based on the evaluation of: - Originality and complexity of the design - Achieved results with respect to simulation and prototyping - Submitted design files - Test plan - Final report - Presentation and defense - Collaboration within the group The individual grade is then determined by increasing or decreasing the group grade by a maximum of 1 point, based on a peer-review within the group. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr.ir. A.J. van Genderen	
Co-Instructor	Prof.dr.ir. G. Gaydadjiev	
Co-Instructor	Dr.ir. M.C.R. Fieback	
Co-Instructor	Dr.ing. C.P. Frenkel	
Co-Instructor	C. Gao	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Bachelor Electrical Engineering**

B-EE jaar 3 2024

Director of Studies	Dr.ir. N.P. van der Meijs
Director of Studies	Dr.ir. R.C. Hendriks
Program Coordinator	Dr. B. Hunyadi
Program Coordinator	C. Thepass
Prerequisites	Let op! De Ingangseisen zijn in de OER vastgelegd. Mocht er verschil zijn tussen wat er hier staat en wat in de OER staat, dan is datgene wat in de OER staat geldig. Om deel te mogen nemen aan het studieonderdeel EE3L11 Bachelorafstudeerproject Electrical Engineering moet je alle vakken van het eerste en tweede jaar van het je bachelorprogramma behaald hebben.
Introduction 1	Inschrijven Voor EE3L11 Bachelorafstudeerproject Electrical Engineering dien je je in te schrijven. Meer informatie vind krijg je via een announcement via Brightspace en tijdens de informatiebijeenkomst in het 3e kwartaal. Inschrijven Voor de (deel-)tentamens van álle vakken moet je je uiterlijk 14 dagen vóór de dag van het tentamen via OSIRIS inschrijven. Tussen de 13 en 6 dagen voor de tentamendag is het mogelijk om een late inschrijving te doen. Je komt dan terecht op de wachtlijst en kan deelnemen aan het tentamen als daar plek voor is. Wil je bij nader inzien toch niet meedoen aan een studieonderdeel of tentamen waar je je voor hebt ingeschreven? Schrijf je dan ook weer uit! Zie voor uitleg https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens
Introduction 2	In het derde jaar volg je een minor. Kijk voor meer informatie op http://minors.tudelft.nl. Als je na je bachelor nog een master wilt volgen, dan is het nu tijd om je te oriënteren. Er wordt twee keer per jaar een masterevent georganiseerd door de TU Delft. Tevens kun je kijken op doorstroommatrix.nl voor welke opleidingen je toelaatbaar bent.

EE3C11	Electronics	5

Responsible Instructor Dr.ir. C.J.M. Verhoeven

Contact Hours / Week x/x/x/x 0/0/6/0 hc

Education Period 3

Start Education 3

Exam Period 3
5

Course Language English

Course Contents

This course is divided into two sections: Electronics and Device Physics.

The Electronics section focuses on the structured design of analog electronic circuits. The design methodology taught in this course is grounded in network theory, control theory, signal processing, and physics.

Designing analog electronic circuits is often considered complex due to the numerous performance aspects that can be achieved through various methods. Designers face many degrees of freedom to obtain the desired circuit performance, making the design space vast and potentially confusing. As a result, analog electronics is sometimes seen more as an art than a solid discipline.

Rather than starting with numerous existing solutions to known problems, this course demonstrates the effectiveness of beginning a new design by clearly distinguishing theoretical concepts and studying their optimal implementation. This approach leads to clear, reliable designs with predictable performance and fosters knowledge sharing and development.

During the design process, designers must find real-world components, determine their operating conditions, and fix these conditions through design. This involves formulating and solving design equations for the circuit. For instance, determining a budget for the gain-bandwidth product of an operational amplifier requires establishing relationships between this product and the performance requirements of the circuit incorporating the amplifier.

Setting up design equations for analog circuits, even those based on operational amplifiers, is challenging and requires knowledge of technology, modeling techniques, and the interactions between circuit behavior and operational amplifier properties.

Although this method applies to various electronic systems, this course emphasizes the design and implementation of high-performance operational amplifier circuits. (Transistor-level details will be covered in the master course EE4109.) Throughout the course, the design of a hearing loop systemcomprising a transmission loop antenna, a receiver coil, and the transmitter and receiver electronicsserves as a practical example, with theory applied to this system in each lecture.

The Device Physics section focuses on the physical description of the most relevant transistors, including bipolar and MOS-based transistors.

Device models are crucial for translating an ideal electrical network into an electronic circuit. These models include parts that match the required ideal network theoretical behavior and parts that model physical limitations. Electronic design involves finding the optimal electrical network to meet system requirements and then translating it into a physically realizable network by refining the models and minimizing the influence of introduced non-idealities.

Professional designers need to understand the physical background of device models to estimate parameter accuracy and stability. This knowledge is essential for predicting circuit performance in the real world and identifying areas for technological improvement. Recognizing which parameters most significantly restrict circuit performance is crucial for selecting the best technology to meet requirements or demonstrating that the requirements exceed current physical limits.

Study Goals

Upon completing the course, you will be able to:

Translate an application description of an amplifier into a functional specification, including performance metrics, operational conditions, and cost considerations.

Design appropriate amplifier concepts based on source and load specifications.

Identify performance factors such as noise, signal handling capacity, and bandwidth of basic operational amplifiers.

Select operational amplifiers with suitable specifications for the intended design and optimize their operating conditions to affect these performance factors.

Design operational amplifier-based amplifiers, including high-performance variants, using error reduction techniques such as compensation, balancing, and feedback.

Establish design considerations for noise and make informed decisions during the design process.

Establish design considerations for signal handling capability and power efficiency, and implement appropriate design choices.

Establish design considerations for frequency response, distortion minimization, and biasing, and implement corresponding design strategies.

Design and implement proper biasing circuitry based on established design considerations.

Evaluate the performance of the final design against specified requirements and suggest improvements where necessary.

Select appropriate models and design verification tools for each stage of the design process.

Compile a comprehensive design report that justifies all design steps, evaluates performance, and provides clear explanations.

Utilize fundamental semiconductor physics concepts to explain the operation of PN junctions.

Explain the characteristics of bipolar and CMOS transistors using corresponding electrical models derived from physical principles within the transistors.

Education Method

The main format for the Device Physics section is Lectures and Colstructions.

The electronics section of the course is structured around the concept of Object-Oriented Education. Here, the object is a hearing

	loop system that transmits audio through a large loop antenna to a receiver with a small reception coil. From the beginning, theoretical concepts are systematically presented to be directly applicable to characterizing the antenna system and designing the transmitter and receiver circuits. Students are invited to form design teams of approximately six members to collaborate on the design throughout and beyond the lectures. Students opting to work on the design as a team will engage in a poster session for peer review at the end of the lecture series. This session determines whether the designs are ready to be built and tested in the Tellegen Hall. The practical session in the Tellegen Hall concludes with a team design review, which may earn a bonus grade for the Electronics part of the course. If this bonus grade is higher, it can replace the exam grade for the Electronics section. This approach is considered the most effective way to learn systematic electronic circuit design. While collaboration is recommended, studying the electronics part individually is entirely feasible, including online through the companion website.
Computer Use	During the design creation phase, designers must identify real-world components, determine their operating conditions, and establish these conditions through design. This involves formulating and solving design equations for the circuit. For example, to determine a budget for the gain-bandwidth product of an operational amplifier, we must identify the relationships between this product and the performance requirements of the circuit containing the operational amplifier. Setting up design equations for analog circuits, even those using operational amplifiers is challenging and complex. It requires knowledge of technology, modeling techniques, and the interactions between the circuit's behavior and the operational amplifier's properties. This process is closely linked with symbolic analysis. To generate the symbolic equations for the designs created during the course, the Symbolic Linear Circuit Analysis Program (SLiCAP) is used. Students can refer to the examples and perform SLiCAP simulations to understand circuit behavior. However, using SLiCAP is not mandatory for studying this course. SLiCAP can be downloaded from: https://analog-electronics.tudelft.nl/SLiCAP.html.
Assessment	The exam consists of two sections: Electronics and Device Physics. It is taken through two separate individualized quizzes on Brightspace, making enrollment in Brightspace essential for participation. Students who previously took the course must also enroll in this year's Brightspace site. The final exam grade is the average of the grades from the Electronics and Device Physics sections. For students who received a bonus grade during the design review, this bonus grade will replace the Electronics section grade if it is higher. The bonus grade is also valid for the re-sit exam. However, the individual grades of the two sections do not remain valid for the re-sit exam. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Co-Instructor	Dr. R.A.C.M.M. van Swaaij
Co-Instructor	A.J.M. Montagne

EE3D11	Computer Architecture and Organisation	5
Responsible Instructor	Prof.dr.ir. S. Hamdioui	
Instructor	Dr.ir. M. Taouil	
Instructor	Dr.ing. M. Shahraki Moghaddam	
Instructor	Ing. A.M.J. Slats	
Contact Hours / Week x/x/x/x	0/0/4/0 hc; 0/0/4/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Expected prior knowledge	Digital systems or equivalent course on logic design Basic concepts of programming in C	
Course Contents	This course provides an overview of the architecture and organization of a computer hardware system and the important principles of computer organization. In addition, the course addresses also object-oriented programming techniques. The course demonstrates the interrelation between hardware and software, and illustrates how the computers operates and how they can be programmed, with the emphasis on processor design and implementation. Topics discussed are Computer system overview, Measuring and comparing performance, ISA: instruction set architecture (MIPS, x86, 8051, JVM), Computer arithmetic, processor implementation, fast processor implementation, Memory hierarchy and caching, Interfacing, Modern architectures and organizations (superscalar, VLIW/IA64/Itanium), etc.	
Study Goals	The main aims of the course are: 1. Understand the basics of computer abstraction and performance: what are the different computers? How to evaluate performance? What is the impact of technology scaling? Etc. 2. Understand and use the computer language: What is an ISA and how to write and extend it? How does it support procedures in hardware? Impact of language selection of hardware design? Etc. 3. Understand and perform computer arithmetic: how to perform addition, subtraction, multiplication and division for integers and floating point? What is the impact of the algorithm on hardware implementation? Etc. 4. Understand and use the principle and techniques used to design a processor: how does hardware execute instructions? What are the performance characteristics and limitations? How to accelerate the performance? Etc. 5. Understand the memory hierarchy and its impact on performance: What is the memory hierarchy? What are the characteristics and limitations of each hierarchy level? What are the different alternatives to improve the performance? Etc. 6. I/O systems and their interaction with the processor: what are the different I/O? How does the communication take place? How are the dependability, reliability and availability requirements for each I/O? etc. 7. Understand and describes the trends and the challenges in the field of computer architecture. 8. Use object-oriented programming techniques in C++ such as: classes and objects (definition and implementation) and inheritance (types, multiple inheritance, redefinition).	
Education Method	Lectures + lab	
Literature and Study Materials	Lecture Notes Computer Organization & Design MIPS Edition: The Hardware/Software Interface, by David A. Patterson &John L. Hennessy; 6th Edition; Morgan Kaufmann E-book "C++ Language Tutorial," Juan Soulie, www.cplusplus.com/doc/tutorial	
Assessment	The assessment consists of the following parts: 7 Lab assignments: Pass/Fail 6 homework: HW1, HW2, HW3, HW4, HW5, and HW6 (weekly). 1 written exam: WE Final grade of the first exam If [(Lab passed) AND(HW>=6)] then: 'Finale grade' = max((0.20*HW + 0.80*WE), WE) Where HW is the weighted average grade of all homework; HW = (HW1 + HW2 + HW3 + 2*HW4 + 2*HW5 +HW6)/ 8. WE is the grade of the written exam. Final grade of retake exam: If (Lab passed) then: 'Finale grade' = WE Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Judgement		
Co-Instructor	Dr.ir. M. Taouil	
Co-Instructor	Dr.ing. M. Shahraki Moghaddam	

EE3L11	Bachelor graduation project Electrical Engineering	15
Responsible Instructor	Dr.ing. I.E. Lager	
Instructor	Dr. J. Hutton	
Instructor	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	0/0/0/5 project (0/5/0/0 project)	
Education Period	1 2 4	
Start Education	1 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Access to the Bachelor Graduation project (BAP) is granted upon fully completing: (i) the curriculum for the first two years for access to the BAP in Q4 (main entry point); (ii) the complete Bachelor curriculum for access to the BAP Q1/2 (alternative entry point). The reference moments are: (i) the end of the first semester for the main entry point; (ii) the beginning of the first semester for the alternative entry point.	
Course Contents	The Bachelor graduation project (BAP) is a complex educational activity offering the students the possibility to materialise the knowledge and practical skills accumulated during the complete Bachelor curriculum in a design project in the area of Electrical Engineering. The students are expected to go through the complete design process, from drawing up a set of requirements, through the concept analysis and design, concept selection, to the presentation of the final design, in a structured and iterative manner. For stimulating an entrepreneurial spirit, the graduation project also requires the students to write a business plan describing how the technology can be transformed into some commercial product. In view of the prime importance of the societal awareness for the future engineers, the students will be also involved in an Ethics and technology module that will discuss responsibility, ethical theories, ethical dilemmas in companies, taking decisions in an environment of uncertainty, ethics and law etc. The modules expected effect is to train students to think about their role as future engineers and be able to raise their voice and participate in moral debates in their future occupation. As a rule, the project is to be carried out in groups of 6 students. However, projects effectuated in groups of 4 or 2 will also be possible the suitability of such projects will be examined on a case-by-case basis. Individual projects are not accepted. The groups task is approached concurrently by 3 or 2 subgroups that address specific aspects or components of the given assignment in the case of projects effectuated by groups of 2, no subdivision will be applied. Each subgroup is required to draw a thesis (one per subgroup) that must then be defended in front of a jury consisting of specialists in the area (see below). The complete group must also jointly prepare one Business Plan that must be presented and defended during a competition termed Grand Finale. The BAP is carried out, as a rule, in Q4 of the third year (the main entry point). Access to this discipline is only granted upon fully completing the curriculum for the first two years. Alternatively, the BAP can also be effectuated in Q1/2 (the alternative entry point), in which case the entry requirement is the complete Bachelor curriculum, except for EE3L11. The reference moments are: the end of the first semester for the main entry point; the beginning of the first semester for the alternative entry point. This discipline consists of 3 modules: Bachelor thesis, Business Plan and Ethics and technology. The credits are granted as follows: 10 ECTS for the Bachelor thesis itself, 3 ECTS for the Business Plan module and 2 ECTS for the Ethics and technology module.	
Study Goals	After following this course the student should be able to: Apply, integrate and deepen the electro-technical knowledge accumulated in the previous study years. Account for the societal facets of the electro-technical engineering. Cooperate in a group for resolving a complex, open task. Identify the pertaining relevant literature sources and assess their suitability. Devise and articulate a design process according to the generally accepted procedures. Develop a prototype. Report on a technical subject in a scientific manner that also highlights the development process. Present a technical subject in a clear manner to a professional audience. Experience the various aspects involved by starting up a company. Write a (relatively simple) business plan and present/defend it convincingly. Critically and pro-actively analyse and address the societal facets of industrial activity, with an emphasis on those occurring in the realm of electrical engineering. Recognise moral dilemmas in the vocational practice. Be able to judge moral issues with the instruments of ethical reflection.	
Education Method	Design project in a team. For some skills there will be refresh instructions for which presence is mandatory.	
Literature and Study Materials	All required course material is offered via Brightspace. Recommended textbooks are also indicated on Brightspace and in the course manual (see Practical Guide).	
Practical Guide	EE3L11 Bachelor Graduation Project. Business plan, Ethics and technology, Bachelor graduation thesis available in Brightspace.	
Assessment	Students must submit a Bachelor graduation thesis, one per subgroup. The thesis must be defended per subgroup in front of a graduation commission appointed by the faculty, the assessment of the commission being for each student, individually. The complete group must draw a Business Plan. The plan must be defended per group in front of a commission, the assessment of the commission being for the complete group. The Ethics and technology module is assessed based on the activity during the course and an essay written per subgroup (E&T subgroups do not necessarily overall the project groups/subgroups). In case of an insufficient result for any of the three components, repair options may exist in accordance with Article 2, Examination Requirements, Clause 4, of the Implementation Regulations 2023-2024. These options will differ depending on the specific component, with details being provided in the course manual (see Practical Guide) and/or during the introductory meetings of the relevant component. The Bachelor thesis is assessed based on a exam form + rubrics that are included in the course manual. The Business Plan is assessed based on the submitted business plan and the group results during the Grand Finale. The Ethics and technology is assessed based on the activity during the course and a final essay. The final grade is computed as G = 10/15*Bt + 3/15*Bp + 2/15*E&t where Bt is the partial grade for Bachelor thesis, Bp the one for Business Plan and E&t the one for Ethics and technology. The minimum for all partial grades is 6.0. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE3P11	Electromagnetics	5
Responsible Instructor	Dr.ir. R.F. Remis	
Instructor	O.A. Krasnov	
Contact Hours / Week x/x/x/x	0/0/6/0 hc; 0/0/6/0 prac	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Expected prior knowledge	Advised: EE1M11/21/31 Linear algebra and calculus (differentiation, integration, complex numbers, Euler's formula, vector algebra, vector calculus, gradient, divergence, curl, integral theorems for gradients, divergences (Gauss's integral theorem), and curls (Stokes's integral theorem)), EE1P21 Electricity and Magnetism (electrostatics and magnetostatics), EE1C11/21 Linear Circuits (Kirchhoff's laws, resistance, capacitance, inductance), EE2S11 Signals and Systems (the Fourier transform), EE1C11/21 Complex function theory and differential equations (principle branch of the square root and simple differential equations)	
Course Contents	Review of vector algebra and calculus, electrostatics, magnetostatics, Lorentz's force law, Maxwell's equations in vacuum, the compatibility relations and causality, conservation of charge, time-harmonic electromagnetic fields, polarization state, Maxwell's equations in matter, induced currents and the constitutive relations, the electromagnetic boundary conditions, Poynting's theorem for transient and time-harmonic waves, uniform and nonuniform plane waves, TE and TM electromagnetic waves, reflection and transmission at planar interfaces, the Fresnel reflection and transmission coefficients, Brewster angle and total reflection, the electromagnetic field in a highly conducting material, eddy currents, skin effect, induced surface current and surface impedance, transverse electromagnetic waves (TEM waves), transmission lines and the basic transmission line equations (telegraph equations), characteristic impedance, the coaxial line and the parallel-plate waveguide, propagation along lossless and lossy transmission lines, introduction to radiation, generalization of Coulomb's law and the Biot-Savart law for time-dependent fields	
Study Goals	After a successful completion of this course, you will be familiar with electromagnetic fields and electromagnetic wave propagation in a variety of settings as encountered in Electrical Engineering and other fields. You will have a thorough understanding of Maxwell's equations and you will be able to explain in your own words how electromagnetic waves propagate through different materials. Moreover, you will know and understand the different kinds of reflection and transmission effects that can take place at the interface between two different media. You will also be able to predict what types of induced phenomena (such as heat production, charge accumulation, and surface current generation) can take place within a piece of matter or at its boundaries. In addition, you will have an understanding of how the electromagnetic field can be used to transport energy (information) from one point to another and you will recognize that so-called transmission lines can be used to guide this energy transport. Finally, you will know how the voltages and currents along these lines are related to the electromagnetic field quantities and you will have a firm knowledge and understanding of the propagation properties of these lines.	
Education Method	Lectures and tutorials	
Literature and Study Materials	Book: Fundamentals of Applied Electromagnetics, F.T. Ulaby, E. Michielssen, and U. Ravaioli, 8th edition, Pearson. Slides: Available on Brightspace.	
Assessment	Written exam and lab sessions. In case of insufficient results for the course labs, a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. The course EE3P11 Electromagnetism consists of lectures and two obligatory lab sessions. Your final grade is computed as follows: T= your grade for the exam or resit exam, L = your grade for the lab sessions. If T>=5,0 and L>=6,0 then your final grade = 0.85*T + 0.15*L. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	A one-page summary (handwritten, two sides of one A4) and a (graphic) calculator.	
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